# LOW-RESOURCE NEURAL MACHINE TRANSLATION FOR ENGLISH-TO-IGBO USING RNNS AND TRANSFER LEARNING

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**DIRECTION:** 01.04.02 APPLIED MATHEMATICS AND INFORMATICS

**PROGRAM:** MSc ADVANCED METHODS OF MODERN COMBINATORICS

**GROUP:** M05-017b

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JUNE, 2022

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Thesis Submitted in Partial Fulfillment of the Requirements for the Award of Master of Science (MSc) Degree in Applied Mathematics and Computer Science at the Center for Educational Programs, Phystech School of Applied Mathematics and Informatics(FPMI),Department of Discrete Mathematics.

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# Abstract

The emergence of Neural Machine Translation systems has improved translation quality between language pairs with rich morphology. However, with the enormous success achieved with NMT, little or no research has been done on low-resource and endangered languages.

In this study, we developed a neural machine translation model from English to Igbo, an African language spoken by over 40 million people in Nigeria, Cameroon, Haiti and across west Africa. We used the standard benchmark dataset collected from bible corpora, local news, Wikipedia articles, and common crawl verified by language experts.

In our experiments with the LSTM and GRU RNN architectures, the GRU performed better on short sentences, while the LSTM model performed better on shorter and longer sentences. We further showed that better performance is achieved with the Dot-product attention scoring function. The quality of translation demonstrated by our model matches the Eng-Igbo state-of-the-art benchmark. Our model also outperformed the Totaoba benchmark by a BLEU point of +9 when trained on a simpler structured dataset (English to French).

We also showed the effect of transfer learning on the translation quality. Our transfer-learning model outperformed the English-Igbo state-of-the-art benchmark up to +4.83 BLEU points, achieving 70% translation quality, a notable improvement on a dataset with complex Subject-Verb-Object (SVO) word order.

***Keywords:*** English, Igbo, Machine Translation, RNN, Natural Language Processing

# Acknowledgement

My gratitude first goes to God almighty for His grace and mercy during this research work. I also wish to express my gratitude to my Thesis supervisor, Biswarup Das (PhD), for his guidance and encouragement throughout this work despite his busy schedule. Your close and thorough supervision at every phase of this research gave me the confidence needed to accomplish more tasks needed to meet our research objectives. Your kindness and tolerance allowed me to openly talk to you and seek your advice whenever needed. Sir, you did not only supervise my thesis work, but you also motivated me to acquire top research skills needed for future research and career journey. I am forever grateful for your mentorship.

I would like to appreciate my friend and roommate, Tendai Chikake (PhD Researcher at MIPT), for his encouragement, and kind support. Thank you for always being willing to review my drafts and make the necessary corrections despite your busy schedules. Your positive motivation and eagerness to respond at any time remain indelible in my heart.

I would also like to express my deepest gratitude to the Open Doors Olympiad Russian Scholarship Project for granting me this prestigious opportunity to study at the best technical institution in the Russian Federation (Moscow Institute of Physics and Technology National Research University-MIPT). You did not only fund my studies, but you also exposed me to the cosmopolitan culture in Russia. You have granted me the privilege to interact with the international community, which has broadened my view point about life. I am grateful to the wonderful MIPT team at the international office, social dean, and the Faculty of Applied Mathematics and Computer Science (FPMI) for your generosity and care.

My deepest appreciation goes to my beloved mother, Mrs. Ekle Patricia, my siblings (Ekle Grace, Blessing, James, Peter, and Emmanual), and my extended relatives for their love, support, and encouragement during my program. I will not forget your sacrifice of love. I want to also appreciate the Moscow Good News Church, my friends, and colleagues. In particular, Fakunle Elijah, Isaac ThankGod, and Piyalee Pattanaik.

To my dear friend and brother, Joseph Luper Tsenum, thank you for your kindness, motivation, and continuous support of my academic progress. You're indeed a true friend. To the New Life Research Foundation family, I appreciate your love, prayers, and support in my academic pursuit. May the good Lord, in his infinite mercy, bless you all.

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# Chapter 1

## INTRODUCTION

## 1.1 Background of Study

Machine Translation is a sub-field of computational linguistics and one of the applications of natural language processing (NLP). The importance of language as a communication tool cannot be overemphasized, cutting across different sectors of human interaction, ranging from politics to international trade and tourism. Language variance has raised a number of problems from one language to another, which has motivated much research in the field of NLP. Natural Language Processing is a branch of Artificial Intelligence that uses linguistic and computational techniques to effectively analyze and interpret human language. Linguistics, on the other hand, is the study of language structure, grammar, meaning, syntax, and phonetics. Some of the applications of language processing are machine translation, text summarization, sentiment analysis, question answering, text mining, chatbots, and recommender systems (Ba et al., 2021).

Machine Translation (MT) is the process of using automated machines to translate text from one language to another. MT takes a sentence in one natural language (source input) and translates it into a target sentence in another natural language (target output), where the meaning of both the source and target sentences should be the same. This technique works with substantial amounts of source and target data that are compared and matched against each other by a machine translation engine. Research in machine translation is critical in natural language processing for a variety of reasons. It helps in bridging the communication gap among people from all over the world who speak different languages. And, it drives down the cost of translation, saves time and provides fast services for real-time enterprise translation (He et al., 2019).

Researchers have used a variety of approaches in developing machine translation, which could be grouped into three categories: rules-based machine translation (RBMT), statistical machine translation (SMT), and neural machine translation (NMT) (Costa-Jussa and Fonollosa, 2015). Rule-based systems use expert-developed linguistic knowledge, such as grammar and language rules. SMT does not rely on rules to learn how to translate; instead, it analyzes a vast number of current human translations. While the NMT, on the other hand, uses neural networks to teach itself how to translate, A neural network is a technique in machine learning that tends to mimic the operations of the human brain. It consists of input layers, hidden layers, and an output layer.

Sutskever et al. (2014) describe Neural Machine Translation (NMT) as an advancement in the field of machine translation in which a massive neural network is developed and trained to read a text and provide an accurate translation (Sutskever et al., 2014). NMT uses the encoder-decoder architecture. The method consists of an encoder and a decoder for encoding a source sentence and decoding it to a target sentence. By incorporating neural components into conventional translation systems such as phrase-based systems that translate the entire sequence of words, NMT architectures have demonstrated impressive results when compared to previous language translation techniques. and has come close to achieving state-of-the-art performance (Bahdanau et al., 2014). Concisely, studies show that neural machine translation learns faster and produces more accurate translations compared to other machine translation techniques. Hence, the present demand for translation and translation tools surpasses the capability of available solutions, necessitating increased research in the field of machine translation.

## 1.2 Statement of Problem

NMT research has sparked a lot (Zhou et al., 2016; Schmidt and Marg, 2018). A series of studies have been conducted to investigate the deep learning techniques employed in neural machine learning. Detailed results are compared in terms of decoding, modeling, interpretation, data augmentation, and evaluation (Tan et al., 2020). There are multiple approaches for developing powerful encoder and decoder components of NMT, which include the recurrent neural networks (RNN)-based methods, convolution neural networks (CNN)-based methods, and self-attention network (SAN)-based methods (Stahlberg, 2020). However, RNN-based model is mostly preferred for neural machine translations. An RNN is a supervised machine learning model composed of artificial neurons with one or more feedback loops, in which a parallel corpus is trained to minimize the difference between output and target pairs by optimizing the network's weights (Datta et al., 2020). Furthermore, a fraction of the corpus is used as the validation dataset to keep the network from under-fitting or over-fitting. Unlike Artificial Neural Networks, Recurrent Neural Networks can simulate sequences of data, such as time series, in which each sample is supposed to be dependent on the past iterations. RNN can also be used in conjunction with convolutional layers to significantly boost pixel neighborhood when used in computer vision.

In this project, we use the RNN-based approach to solve the translation task under consideration.

## 1.3 Objective of the Research

The aim and objective of this research is to build a neural machine translation system using the Recurrent Neural network architecture.

The study is set specifically to:

i constructing the RNN-based encoder-decoder model which translates English to Igbo language (spoken by over 40 million people across Nigeria).

ii combine the attention mechanism to enable the decoder to focus on relevant word vectors.

iii experiment with different attention scoring techniques

iv incorporating the concept of teacher-forcing algorithm for training the weights of the RNN and for providing ground truth information to the decoder block.

v implementing the Greedy decoding and Beam Search decoding techniques.

vi obtain the evaluation of our model using the BLEU-score.

vii and to further evaluate the effect of transfer learning on our translation task.

# Chapter 2

## LITERATURE REVIEW

## 2.1 Introduction

In this chapter, we review previous studies on natural language processing and machine translation. We discussed the relevant background of NLP and neural machine translation. Then we identify the Recurrent Neural Network (RNN) architecture and the several types of RNN models, such as the Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU). Furthermore, we present a detailed report on the traditional RNN-based Encoder-Decoder Sequence-to-Sequence (Seq2Seq) model in comparison with our proposed architecture built on the attention mechanism.

In this review, we bring to light some existing research gaps from previous empirical studies of past and contemporary literature. As a result, our research study seeks to solve a selection of these knowledge gaps.

## 2.2 Natural Language Processing

Researches in the field of natural language processing began in the late 1940s as a combination of Artificial Intelligence and linguistics (Nadkarni et al., 2011). Machine Translation was one of the earliest applications of NLP (Liddy, 2001). In 1947 Warren Weaver proposed the idea of cryptography and information theory for language translation inspired by the success of code breaking in World War II (Schwartz, 2018). His system uses dictionary-look ups of words to match and reorder words after translation, this approach fails to capture the lexical ambiguity inherent in human languages.

This major setback inspired many researchers to explore this field. However, the works of

Chomsky in 1957 on Syntactic Structures which introduced the concept of generative grammar profound clearer insight into NLP and Machine Translation (Lees, 1957). This era led to other application of NLP, such as speech recognition, simple heuristic for reasoning and question answering. Between the 1970s and 1990s, there was an evolution in natural language researches such as the use of parsing to understand semantics and discourse models (Kumar, 2011) and the works of Schank and Tesler on language understanding programs which focused on human conceptual knowledge (such as memory and goal) (Schank and Tesler, 1969). From the 1990s till date, the application of probabilistic and data-driven approaches has become prevalent in natural language processing. Major algorithms in NLP such as parsing, part-of-speech-tagging, word embedding and discourse processing incorporates probabilistic models. The technological advancement in Hardwares such as processor speeds and memory has drastically contributed to resolution in speech and language processing (Liddy, 2001).

NLP researchers are now working on next-generation NLP applications that can cope with general text and account for a significant percentage of linguistic variety and ambiguity. The motivation behind NLP is to build system-based computational techniques to effectively learn and represent text data in human-like format. The foundational step in processing language data is the conversion of text from a symbolic, discrete form to a numerical representation.

### 2.2.1 Levels of Natural Language Processing

The presentation of what happens in language processing is expressed at various levels, namely, phonology, morphology, lexical, semantics, syntactic, pragmatic and discourse (Kumar, 2011). NLP applications are quite complex to process because they do not only depend on data, but also on knowledge regarding various linguistic structures. This has resulted in a series of challenges in language processing, such as the ambiguity problem. which has been approached with knowledge-based solutions (Mahesh and Nirenburg, 1996), the FastText approach to determine the ambiguous word in each context (Uslu et al., 2018), and many more. The phonology level tends to identify and interpret linguistic sounds. **Morphology** deals with the composition and nature of words. The **syntactic** level examines how sentences are constructed and the patterns that are permitted in the language.

The **lexicon** level interprets the meaning of individual words, while **semantic** processing focuses on the interactions between word-level meanings in a phrase to identify the various meanings. Whereas the syntax and semantics levels deal with sentence-length units, the **discourse** level works with units of text that are longer than a sentence (Liddy, 2001). And finally, the **pragmatics** level evaluates the concepts retrieved from the text, verifies the interpretation of the semantic analysis, and utilizes the content of the text for clearer understanding.

### 2.2.2 Word Vector Representation

When processing natural language text, most rule-based and statistical algorithms require the transformation of words into vector formats that can capture their meaning and semantic relationships. One of the simplest forms is to represent words as "one-hot" representations (Brownlee, 2017). This approach fails to capture the similarity between words.

In 1957, Firth introduced the concept of representing words by the means of their neighbours. This is among the most widely used statistical natural language processing techniques. A related approach called **"word embedding"** was introduced by Bengio et al. (2000) to learn context vector representation of words and to predict the frequency of words given their context. This was followed by similar work by Collobert and Weston (2008) to compute word vectors using multitask learning and semi-supervised learning to improve the generalization of shared tasks. The uniqueness of unsupervised approaches is that the generated word vectors capture distributional semantics and co-occurrence statistics.

The idea was used by Mikolov et al. (2013) to construct **word2vec** and examine word embeddings, elucidating word embedding training as well as the utilization of pretrained word vectors. There are two model architectures for that, the continuous bag-of-words (**CBOW**) or the continuous **Skip-gram**. In CBOW, the algorithm predicts the current word from a window of surrounding context words. While in Skip-gram, the model uses the current word to predict the surrounding window of context words.

**GloVe**, another variant of pretrained word embeddings, was later introduced by Pennington et al. (2014). And in 2018, Facebook introduced **FastText**, a new model for word embeddings that can capture multiple word senses, sub-word structure, and uncertainty information (Athiwaratkun et al., 2018).

Furthermore, more advanced word embedding models (tunable and pre-trained models) have been developed in recent time, including, **BERT** by Google team, **ELMO** by the Allen Institute for AI and **GPT-2** by OpenAI (Devlin et al., 2018; Ethayarajh, 2019). Word embeddings have proved to be more efficient than bag-of-words models or one-hot-encoding systems, which use sparse vectors of the same length as the vocabulary (Goyal et al., 2018).

Word embedding have grown in importance as one of the most important applications of unsupervised learning. The semantic connections and properties of word vectors have helped in powerful NLP applications such as Neural Machine Translations, information retrieval, and question-answering tasks.

### 2.2.3 Neural Machine Translation

The earliest days of machine translation (MT) could be traced to the works of Warren Weaver of the Rockefeller Foundation in 1946, with relation to the use of cryptography for translation (Hutchins, 2000). This idea was faced with ambiguity challenges. In response to this, Weaver later postulated that ambiguity in text may be resolved by inspecting a window of n-words around the uncertain word (Schmidt and Marg, 2018), which was later used in statistical machine translation (Brown et al., 1990a).

In 1951, Yehoshua Bar-Hillel, a professor at the Massachusetts Institute of Technology (MIT), became the world's first full-time machine translation researcher (Hutchins, 1997). This was followed by the invention of the automated translator in 1954 by Professor Michael Zarechnak and other researchers at Georgetown University in partnership with IBM, which translates more than sixty (60) Russian words into English (Chéragui, 2012).

In the 1960s, there was a dramatic increase in MT research with the invention of parsing to understand grammar and the birth of computational linguistics, traceable to the impact of David G. Hays seminars at the Rand Corporation in Los Angeles. In 1964, the ALPAC (Automatic Language Processing Advisory Group) committee was created by the American government to study the prospects and possibilities of machine translation, but little progress was made in this regard, leading to a setback in MT research.

In the 1970s, a new era of MT emerged with the invention of REVERSO by some Russian researchers, setting a new standard for machine translation (Hoffenberg, 1999), followed by the creation of the SYSTRAN system (Russian-English). In 1976, the WEATHER machine translator was created by researchers at the University of Montreal, and in 1978, a rule-based ATLAS2 system was created by the Japanese.

During the 1970s and 1980s, significant progress was achieved in the field of automated speech recognition (ASR). This inspired an IBM research group, under the guidance of Fred Jelinek, to examine how to apply these statistical approaches to machine translation, a subject largely dominated by rule-based paradigms (Brown et al., 1990b). And over the next five years, the IBM team developed more sophisticated models such as the Statistical Machine Translation (SMT), which learns probabilistic translation rules from a parallel corpus of translated data without the need for linguistic knowledge or human participation.

Over the last decades, researchers have successfully combined rule-based machine translation (RBMT) and SMT techniques to develop more powerful hybrid machine translations (Alshawi et al., 1998; Yamada and Knight, 2001). And the appearance of websites for automatic translation like Google in 2006 (Bahdanau et al., 2014; Costa-Jussa and Fonollosa, 2015).

### 2.2.4 Machine Translation Approaches

Past NLP researchers have employed a variety of approaches to build machine translation systems. These include the Rule-based Machine Translation (RBMT), Corpus-based Machine Translation (CBMT), and Hybrid Machine Translation (HMT) approaches (Okpor, 2014). The RBMT, also known as the knowledge-based MT, is based on linguistic information about the source and target languages retrieved from bilingual and grammar, using the semantic, morphological, and syntactic rules of both languages. The rule-based approach is split into three (3) sub-approaches; the Direct MT, Transfer-Based MT, and the Interlingual MT approaches.

The corpus-based MT uses large amounts of raw data in the form of parallel corpuses. It is categorized into Statistical Machine Translation (SMT) and Example-based Machine Translation (EBMT). SMT approaches are built on the basis of statistical models such as the Bayes Theorem (Brown et al., 1990b). The EBMT uses the idea of translation by analogy.

Finally, the Hybrid MT approach is simply a combination of the rule-based and statistical approaches. This has proven to have better efficiency. The Neural Machine Translation (NMT) is an advanced version of the Hybrid MT which combines the RBMT, SMT, and deep learning techniques. NMT made rapid progress in recent years, in 2017 Google team announced that its translation services have been replaced by NMT over its previous SMT. Also in 2018, the Microsoft team claimed to have reached a historical milestone in translation (Linn, 2018). Furthermore, research shows that NMT learns faster and provides more accurate translations than other approaches.

## 2.3 Recurrent Neural Networks (RNN)

The concept of recurrent neural networks (RNN) architecture can be traced as far back as the works of Rumelhart and Hinton (1986) to learn strings of characters and the invention of the Hopfield networks by John Hopfield in 1982. This network served as a memory system for addressing continuous data. The motivation behind RNNs is that they are designed to capture temporal dependencies (dependencies that change over time), such as sequential or continuous data. RNNs use state variables to store previous information alongside current inputs to ascertain current outputs. The Recurrent networks have been applied in diverse tasks from the 1990s such as financial predictions and time series analysis (Giles et al., 1997), forecasting of electric loads (Costa et al., 1999), there are being applied in the works of Coulibaly to track water qualities and to forecast the additives need to filter water (Medsker and Jain, 1999). The RNN architecture comes in many variants. The earlier models are the Vanilla (Fully Recurrent Neural Networks) and Simple RNNs (also known as the Elman Network and Jordan Networks). But these networks were found to have vanishing or exploding gradient problems (the difficulties of capturing a long sequence of information over some time 't'), resulting in the discovery of the Long Short-Term Memory cells (LSTM) with impressive results for sequence-to-sequence applications and the invention of the Gated Recurrent Units in 2014 (Heck and Salem, 2017). Another variant of the RNNs is the Continuous Time Recurrent Neural Networks (CTRNN), which are applied in robotics and the modeling of adaptive behaviors. CTRNN uses the system of ordinary differential equations to model incoming inputs (Beer, 1997).

### 2.3.1 Vanishing and Exploding Gradient Challenge

Just like other neural networks, RNNs use back-propagation techniques to update the weight of the network. This happens when the network reverses its steps to change the weights to obtain

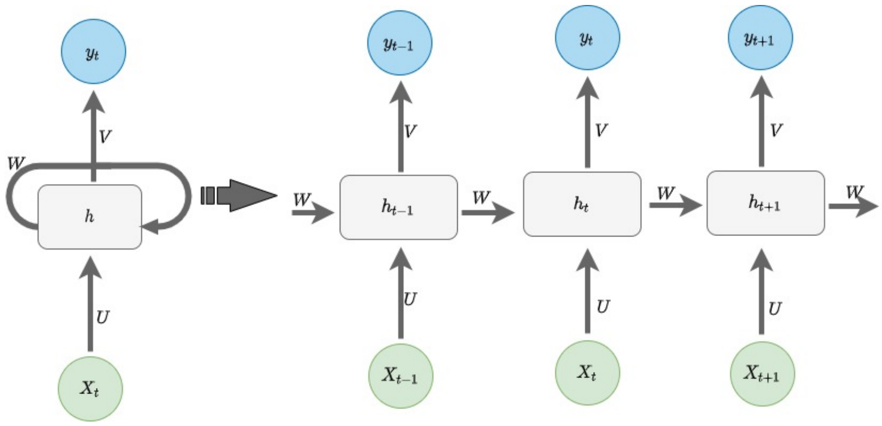



Figure 1: **Vanilla RNN** ($U, V, W$ are the weight matrices of the network)

a better result. When propagating to lower layers, the gradient often gets smaller at each step, leading to a **"vanishing gradient problem"** (Debabala S., 2019) . On the other hand, an **exploding gradient** occurs if bigger weights are assigned and the final output grows infinitely, resulting in an exponential effect on the gradient of the loss. Both these challenges make the RNN models very sensitive to the number of time steps (Goyal et al., 2018). To tackle this problem, a specific type of hidden layer known as the LSTM and GRU model was created.

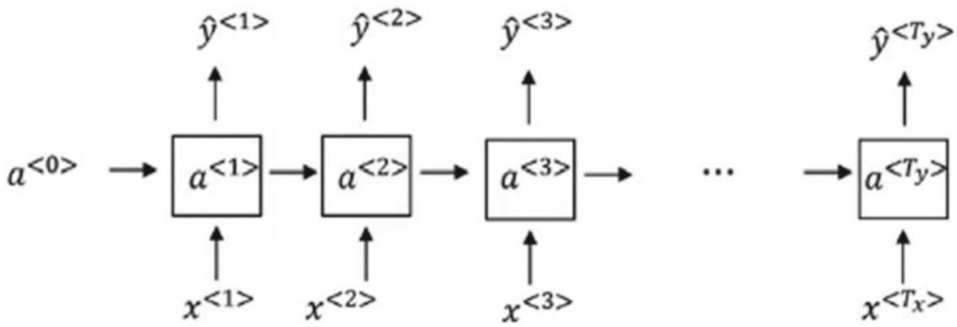



Figure 2: **Vanishing Gradient:** Output $y^{<t_y>}$ is not affected by input sequence like $x^{<1>}$ (Debabala S., 2019).

### 2.3.2 Long Short-Term Memory (LSTM) Model

The Long Short-Term Memory model, which is a variant of RNN, was introduced by Hochreiter and Schmidhuber to rectify the problems faced by the traditional RNN models (Hochreiter and Schmidhuber, 1997). LSTM models help solve the vanishing gradient problems. It is augmented with some recurrent gates called input, forget, and output gates (Gers et al., 2002) and the cell states, which take inputs at every time step. The LSTM prevents back propagation errors from exploding or vanishing gradients, and as a baseline, they are capable of learning long-term dependences and retaining knowledge for a long time Le et al. (2019). According to Schmidhuber (2015), LSTM can be trained to perform tasks that require memories of incidents that unfolded thousands of discrete time steps. In recent times, the LSTM model has won several international pattern competitions and set numerous benchmark records on large and complex datasets (Schmidhuber, 2015). By stacking and training LSTM by connectionist temporal classification (CTC) (Graves et al., 2006), they have been used to solve complex real-world sequence learning tasks such as handwritten recognition and speech recognition. The structure of a traditional LSTM neural network is shown in Figure 9.

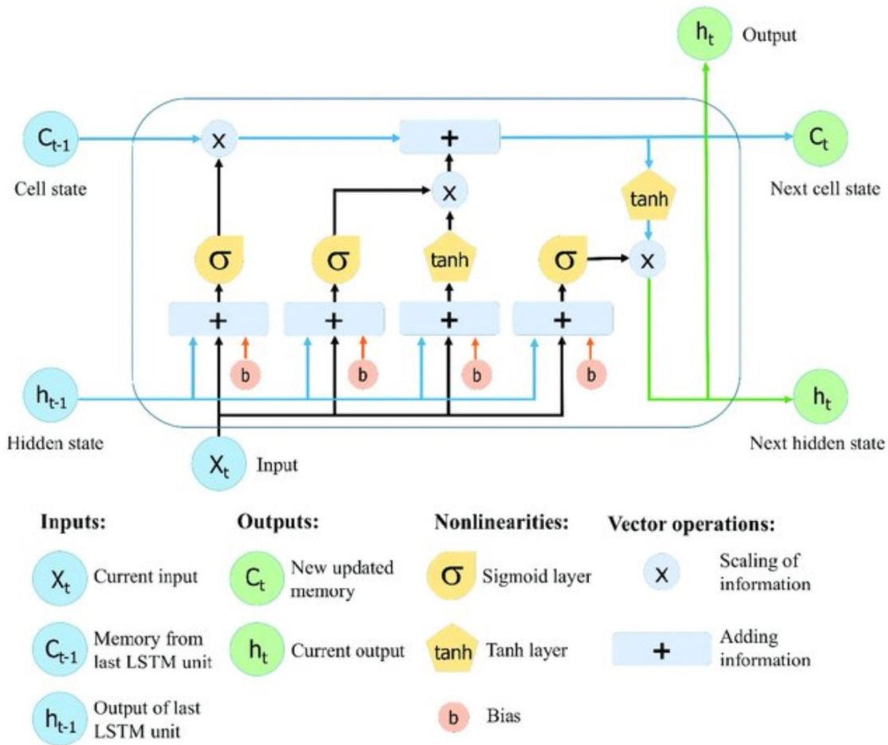



Figure 3: Diagram of the Long Short-Term Memory (LSTM) model. (Zhao et al., 2019).

The basic unit of an LSTM network is the memory block called ”cell states” (containing one or more cell states) (Gers et al., 2002) . The cell state and the hidden state are both passed to the next cell (Le et al., 2019). The main chain of flow is the cell state. It allows for the forward flow of unchanged data.

From Figure 9; $C_i$ is the cell state which is continuous over all time steps and changes as the result of interactions at each time step. There are three gates in an LSTM that help to retain information flowing via the cell state.

**Forget Gate ($f_t$):** This gate controls the limit to which an input is relevant in the memory. It identifies information that is not required and omits it via the sigmoid activation function ($\sigma$) which determines which parts of the previous output to be eliminated.

$$f_t = \sigma(W_f[h_{t-1}, X_t] + b_f)$$

Where $\sigma$ denotes the sigmoid function, $b_f$ is the bias, and $W_f$ denotes the weight matrices. $h_{t-1}$ takes the output of the last LSTM unit ($h_{t-1}$) at time $t-1$ and $X_t$ the current input at time $t$.

**Input Gate ($i_t$):** controls the contribution from the new input to the memory.

$$i_t = \sigma(W_i[h_{t-1}, X_t] + b_i)$$
$$N_t = \tanh(W_n[h_{t-1}, X_t] + b_n)$$
$$C_t = C_{t-1} \cdot f_t + N_t \cdot i_t$$

Here, $C_{t-1}$ and $C_t$ denotes the cell states a time $t-1$ and $t$, $f_t$ is the forget gate at time $t$. While $N_t$ is the candidate value to be added to the input gate output at time step $t$. $(b_i, b_n)$ and $(W_i, W_n)$ refers to the bias and weights of the cell state respectively.

**Output Gate ($O_t$):** this gate determines the value of the next hidden state. It stores information from the preceding inputs. A sigmoid function is used to determine which value of the cell state is outputted. And the output value ($O_t$) is multiplied by the tanh layer of the cell state ($C_t$), with a value of scale at interval $-1$ and 1.

$$O_t = \sigma(W_0[h_{t-1}, X_t] + b_0)$$
$$h_t. = O_t \tanh(C_t)$$

Here $O_t$ denote the output gate output at time step $t$ and $W_0$ and $b_0$ denotes the weight and bias of the output gate at time step $t$.

### 2.3.3 Gated Recurrent Unit (GRU) Model

GRU is a variant of the LSTM model that was introduced in 2014 by Cho et al. (2014). The GRU is a simlified vsersion of the LSTM with two gates (Update and Reset gates) and fewer parameters (Heck and Salem, 2017).The **update gate ($z_t$)** decides whether or not the hidden state should be updated with the hidden state, where as the **reset gate** ($r_t$) indicates whether a particular hidden state should be ignored (Zhao et al., 2019). This model have proven to achieve comparable performance for some NLP applications (Zhou et al., 2016). The operations of GRU can be expressed in the equation below.

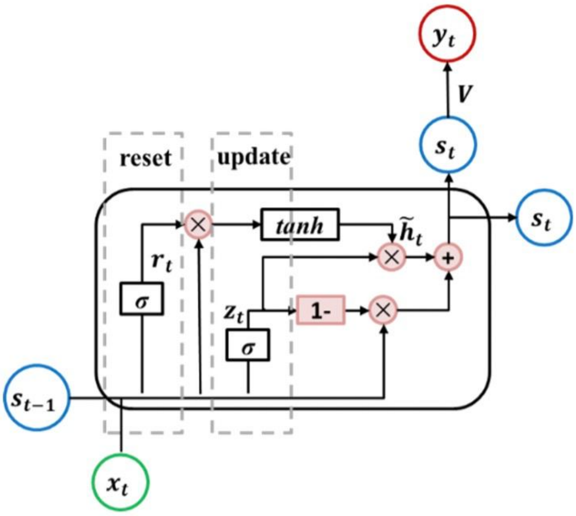



Figure 4: Diagram of the gated recurrent unit RNN (GRU RNN) unit. (Zhao et al., 2019)

$$z_t = \sigma(W_z \cdot [h_{t-1}, X_t] + b_z)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, X_t] + b_r)$$

$$h_t^{`} = \tanh(W_h \cdot [r_{t-1}, X_t] + b_h)$$

$$h_t = (1 - z_t) \cdot h_{t-1} + z_t + h_t^{`}$$

Here, $z_t$ and $r_t$ are the update and reset gate respectively. $W_z, W_r, W_h$ are weight matrices and $b_z, b_r, b_h$ are the bias vectors.

## 2.4 Sequence to Sequence Models

The term "sequence to sequence" or "seq2seq" refers to a larger group of systems that transfer one sequence of data to another. Many of the advanced systems we utilize daily are based on seq2seq models. For example, the seq2seq paradigm is used to power NLP applications such as Google Translate, voice-enabled gadgets, and online chatbots (Sutskever et al., 2014). One of the most used seq2seq models is the "encoder-decoder" architecture.

### 2.4.1 Encoder-Decoder Using LSTMs

As the name implies, the model has two main components called the encoder and the decoder. The basic idea is that the first neural network (RNN) encrypts, or encodes, the information contained in the input sequences as vectors (the hidden state), and passes it to the second neural network (RNN) to predict, or decode, the information contained in the target sentence.

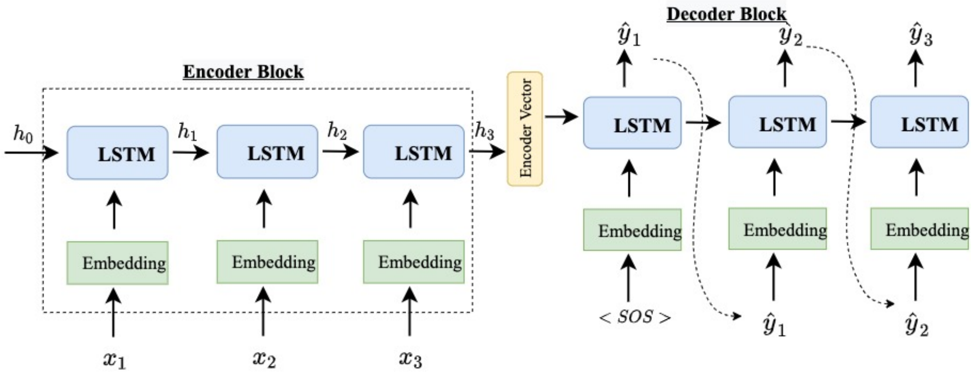



Figure 5: Computational graph of the Encoder-decoder model (Sutskever et al., 2014; Cho et al., 2014)

The **encoder** is constructed with an embedding layer and the LSTM layers. Given some input sequences $(x_1, x_2, \ldots, x_t)$, we look up the embedding and calculate the encoder hidden state $h_t$ at time $t$. We start by initializing the first hidden state, $h_0 = 0$. And at $h_t$ the encoder has seen all the words in the source sentence. $h_t$ is calculated using the formula.

$$h_t = f(W^{(hh)} h_{t-1} + W^{(hx)} x_t)$$

The final hidden state is called the **encoder vector**. This victor encapsulates the information from the entire input sentences and encodes its overall meaning. The **decoder** block is constructed similarly with an embedding layer and the LSTM layers. We us the output word of a step as the input for the next step, that is, $y_t$ at each time step $t$. We start the decoder input sequence with the start of a sentence token (<SOS>) and $\hat{y}_1$ is used as the next input to generate $\hat{y}_2$. Any hidden state $h_t$ is computed using the formula.

$$h_t = f(W^{(hh)} h_{t-1})$$

And our final output $y_t$ at each time step is calculated using the Softmax function. And our final output $y_t$ at each time step is calculated using the Softmax function.

$$y_t = \text{Softmax}(W^s h_t)$$

Since the encoder and decoder in this model are jointly trained to maximize the conditional probability of a target sequence given a source sequence, one major limitation with the framework is that a neural network needs to be able to compress all the necessary information from a source input into a fixed-length vector. This results in the information bottleneck problem, and the model tends to drop off performance on very long sequences.

Ilya Sutskever and colleagues proposed a reversed bidirectional encoder that attempts to solve this problem by reducing the length of dependencies for a subset of the words in the sentence, specifically the ones at the beginning of the sentences (Sutskever et al., 2014). However, the method of reversing the encoder assumes that the order of words in the input and output sequences is similar. This is only possible for languages with similar "subject-verb-object (SVO)" word orders as English and French, but it may not be true for languages with more typologically diverse word orders.

The CNN models were also implemented by some researchers: Bradbury et al. (2016), who incorporated recurrent pooling across a series of convolutional layers, and Kalchbrenner et al. (2016), using convolutional neural networks for sequence modeling (Gehring et al., 2017; Dakwale and Monz, 2017; Khan et al., 2018). However, none of these methods has been shown to be more effective on large datasets.

The Connectionist Sequence classification (CTC) is another popular technique for mapping seq2seq (Graves et al., 2006). The Tree-Structure network is another method in which combined information from each word is guided by some sort of syntactic structure of a tree sentence.

Another method that is widely used in encoder-decoders for translation is the "Emsembling multiple" models. This is the combination of predictions of multiple independently trained models to improve the overall prediction results (Neubig, 2017).

### 2.4.2 Advanced Sequence-to-Sequence Models

The **information bottleneck** problem can be alleviated by adding an attention mechanism into the encoder-decoder framework. This was introduced by Bahdanau et al. (2014). The idea behind attention is that rather than learning a single vector representation for each sentence, we keep all the vectors for every input sentence and refer to them at each decoder step. The attention vector focuses on a particular source word at each given time step. Aside from RNN

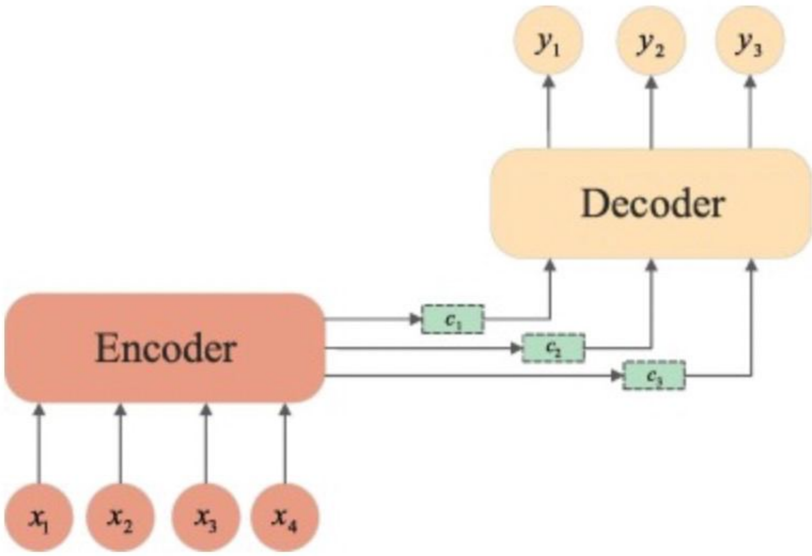



Figure 6: Computational graph of the Encoder-decoder model with Attention mechanism (Bahdanau et al., 2014)

models, other neural networks have been combined with the attention mechanism. In 2017, Gehring et al. (2017) combined the CNN with an attention mechanism. This model enabled the network to capture long-distance dependencies by stacking multiple CNN layers (Gehring et al., 2017). Furthermore, Vasmani et al. proposed the famous Transformer architecture in the paper "Attention is all you need." This model relies totally on the self-attention mechanism to capture its input and output representation. The transformer model consists of the position-wise feed-forward network (FFN) and the multi-head attention layer (Vaswani et al., 2017). It models parallelization by utilizing multi-head attention to focus on information from different representation subspaces from various positions by stacking multiple-attention layers just like the CNN layers (Niu et al., 2021).

The transformer framework has been used to build many state-of-the-art models, such as GPT-2 (Generative Pre-training for Transformer), BERT (Bidirectional Encoder Representations from Transformers), and T5 (Text-to-text Transfer Transformer), which are beyond the scope of our study.

Therefore, this research focuses on building a recurrent neural network-based encoder-decoder architecture paired with an attention mechanism for English to African language machine translation.

### 2.4.3 Summary and Knowledge Gaps

The typical encoder-decoder design by Cho et al. and Sutskever et al. has two flaws that previous seq2seq models for machine translation had to deal with: **First**, there is the long-distance dependency between words that must be translated into one another. This was fixed to some extent in the work of Gusev and Oboturov, by reversing the encoder's direction to kickstart training still, a large number of long-distance dependencies remain, and it is difficult to predict whether we will learn to handle them effectively.

The **second** challenge of the encoder-decoder feature is that it attempts to store all the source sentences of any length in a hidden vector of fixed size, hence leading to an information bottleneck problem. This problem was addressed by implementing the attention mechanism. Afterwards, several works have been proposed as an improvement to sentence alignment and translation performance, the local attention by Luong et al., self-attention networks by Yang et al. (2019). But little or no work has been done to capture languages with scarce datasets, especially African languages.

In this research, we build a NMT model that translates English to the Igbo language (an African language spoken by over 40 million people across Nigeria) using the RNN-based encoder-decoder. In addition to the attention mechanism proposed by Bahdanau et al. and Luong et al., we incorporate the teacher-forcing algorithm to provide ground truth information to the decoder layer. Additionaly, we implements transfer learning techniques to obtain optimal translation quality (Zoph et al., 2016; Nair et al., 2022; Junczys-Dowmunt et al., 2018).

# Chapter 3

## RESEARCH METHODOLOGY

## 3.1 Data Collection

Approximately 12000 Twelve thousand English-Igbo parallel corpus was extracted from the Github repository of Ezeani et al. (2020) (A developing standard benchmark dataset for Igbo-English translation researches collected from Bible corpus, local news, Wikipedia, common crawl, and local materials). The developed parallel corpus was processed and verified by language experts. In addition, we also demonstrate empirically on the Eng-Fra dataset (from the compilation of Tatoaba) to verify the performance of our model.

## 3.2 Text Processing and Analysis

Data pre-processing of the dataset is carried out in three stages: data loading (including visualization and analysis), tokenization, and vocabulary building.

### 3.2.1 Data Loading and Sampling

The parallel corpus is loaded into memory as text data. We placed every English sentence concurrently with the Igbo translation, split by a TAB. We further loaded the parallel corpus into a pandas DataFrame for proper visualization.

### 3.2.2 Sentence Length Analysis

We perform exploratory data analysis to investigate patterns associated with the sentence lengths of the languages. Using the Violin plot and Distplot, the parallel corpus tends to have

| | English | Igbo |
|---|---|---|
| **0** | The news that will interest you: | Akụkọ ndị ga-amasị gị: |
| **1** | Joseph Achuzie, the Biafran brave man is gone. | Joseph Achuzie, Dike Biafra alala |
| **2** | Dapchi: Government has not defeated Boko Haram... | Dapchi: Gọọmenti emeribeghị Boko Haram - Massob |
| **3** | Tottenham look forward to lifting the FA cup i... | Tottenham na-ele anya iburu iko FA na mmeri Ro... |
| **4** | Son Heung-min, Fernando Llorente and Kyle Walt... | Son Heung-min, Fernando Llorente na Kyle Walte... |
| **5** | Son Heung-min of Tottenham and his team mates ... | Son Heung-min onye Tottenham na ndị otu ya na-... |
| **6** | Tottenham has won eight FA cup competitions. | Tottenham emerila n'asọmpị iko FA ugboro asatọ |

Figure 7: Eglish-Igbo Dataset

a symmetric form and the average sentence length falls between 10 and 40.

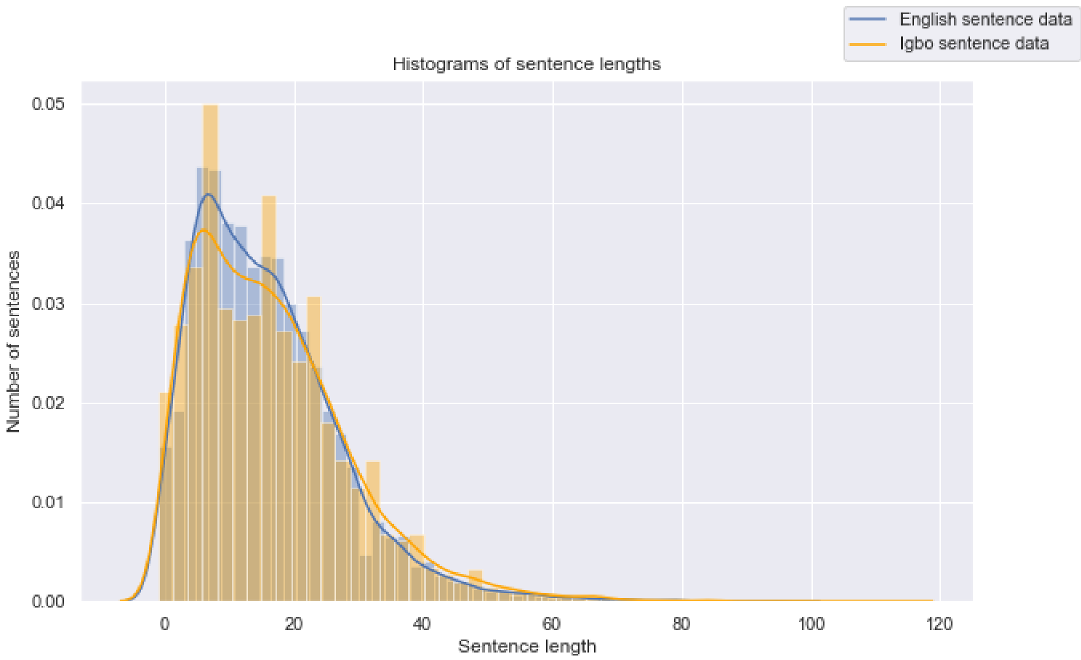



Figure 8: Histogram Plot to Analyse Sentence length

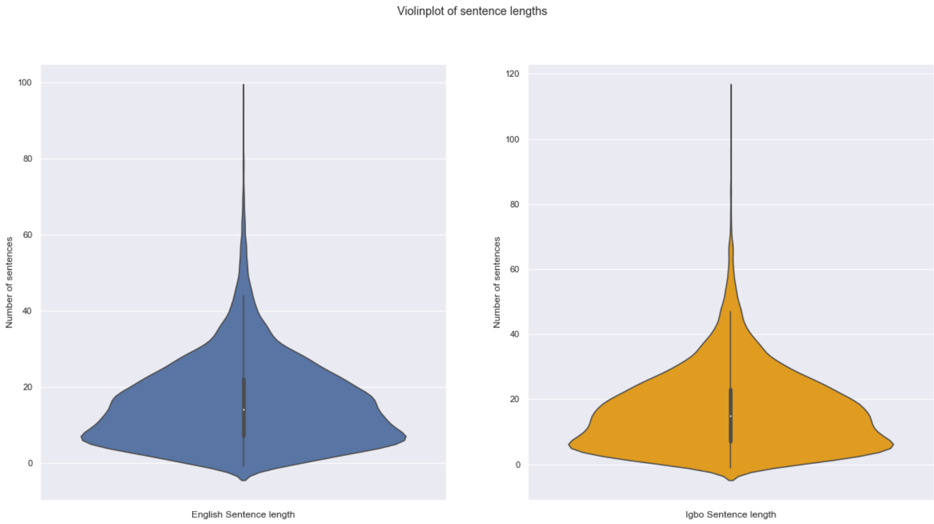



Figure 9: Violin Plot to Analyse Sentence length

### 3.2.3 Tokenization

We used the keras framework Tokenizer to convert words into numerical representation. We performed morphology-based tokenization by filtering off punctuation and numbers, and implemented word splitting.

### 3.2.4 Vocabulary Building

We built 16224 vocabulary from the English corpus and 14,789 Igbo corpus from the Igbo corpus. And we assigned special tokens like " $< pad >$ ", " $< sos >$ " and " $< eos >$ " for padding, at the start of a sentence and end of a sentence, respectively.

| | English Corpus | Igbo Corpus |
|---|---|---|
| Total Words | 176375 | 184538 |
| Vocabulary size | 16224 | 14789 |

Table 3.1: Vocabulary Details

## 3.3 RNN-based Model with Attention

The recurrent neural network (RNN) based model (Bahdanau et al., 2014) implementation contains three major segments; the encoder, the context-vector developed by the attention model, and the decoder layer. The encoder-decoder framework contains two RNNs. One produces a compact representation based on its understanding of the source sentence, while the other generates the equivalent output sentence.

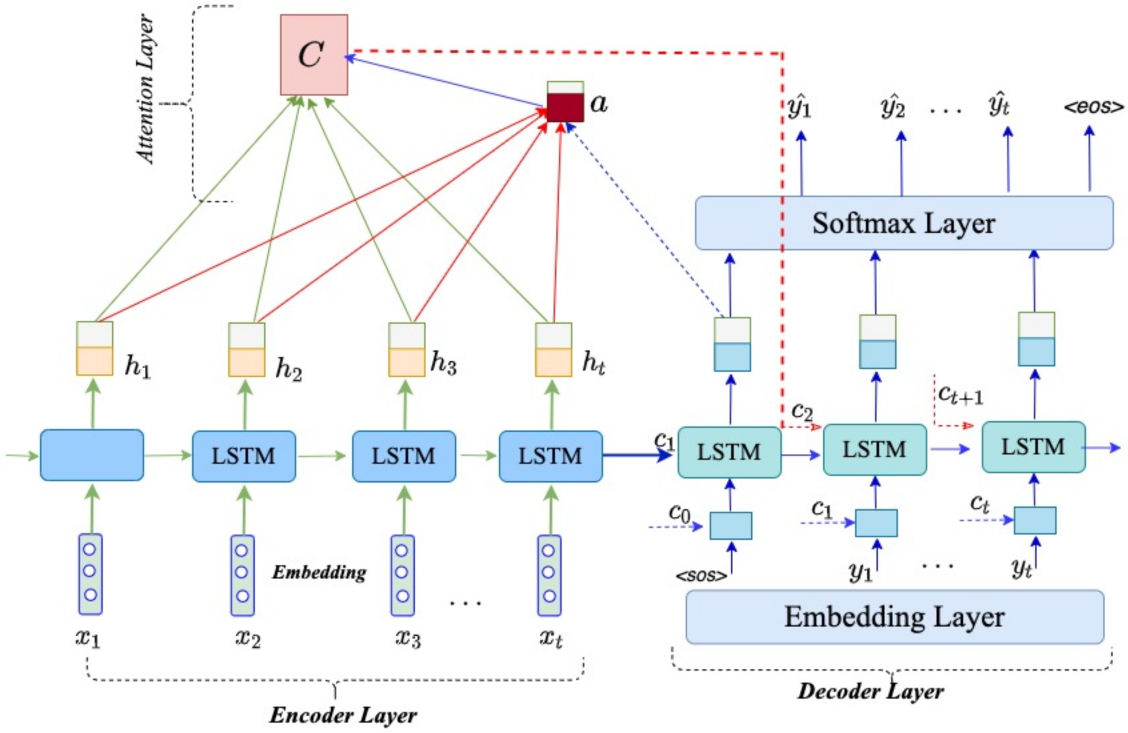



Figure 10: RNN Encoder-Decoder Network with Attention layer.

For model construction and design, I used the Python programming language, and the neural network was developed using the Tensorflow-Keras framework in Python, and I used the Google Colab environment to harness the parallel computational capability of GPUs (NVIDIA Tesla K80" GPUs).

### 3.3.1 Encoder

The encoder block is constructed with a combination of the input embedding layer, the LSTM layer, and the hidden input layers. The embedding layer is used to normalize our input data afterward, the LSTM layer (Hochreiter and Schmidhuber, 1997) is then used to generate a context vector $C$, which represents the entire sequence, that is; at each time step $t$, the hidden state $h_t = \sigma_1(x_t, h_{t-1})$ and finally, $C = \sigma_2(h_1, \ldots, h_t)$, where $\sigma_1$ and $\sigma_2$ are some non-linear functions. The encoder is customized with Batch size 128, embedding 256 and unit of 1024.

### 3.3.2 Decoder

Similar to the encoder block, the layers of the decoder are constructed with the embedding layer, the LSTM model. Following the LSTM, the dense layer is created with the entire vocabulary size. To create the context vector, the Bahdanau attention mechanism is applied. In summary, at each decoder step, attention determines which source input is more important at each decoder phase.

### 3.3.3 Attention method and Context Vector

Our design employs the attention mechanism (Bahdanau et al., 2014). Attention weights are utilized to determine which English words are focused on and which Igbo words are assessed. **The context vector** $C_t$ comprises of the sequence annotations $(h_1, \ldots, h_t)$ to which the encoder, encodes the input sentence where each $h_i$ contains information about the nput sequence with particular emphasis on the part related to the $i^{th}$ word. The context vector $C_t$ at $i^{th}$ decoder time step is calculated as a weighted sum of $h_i$ using the formular:

$$C_t = \sum_{j}^{T_s} \mathbf{a_{tj}} h_j^{\text{encoder}}$$

The weight $\mathbf{a_{tj}}$ for each $h_j^{\text{encoder}}$ is computed by,

$$\mathbf{a_{tj}} = \frac{\exp(e_{tj})}{\sum_{k=1}^{T_s} \exp(e_{tj})}$$

Where,

$$\mathbf{e_{tj}} = \mathbf{atten\_score}(\mathbf{h_{t-1}^{\text{decoder}}}, \mathbf{h_j^{\text{encoder}}})$$

The expression, $e_{tj}$ is an alignment model which scores how well each encoded input matches the current output of the decoder. And **a** is a feed-forward network that is trained in connection with other components. The **scoring function** can be computed using different variants.

$$\text{Score}(h_t, \tilde{h_s}) = \begin{cases} h_t^T \tilde{h_s} & \text{Dot-Product} \quad \text{(Luong et al., 2015)} \\ h_t^T W_a \tilde{h_s} & \text{General} \quad \text{(Luong et al., 2015)} \\ V_a^T \tanh(W_a[h_t^T \tilde{h_s}] & \text{concatenation} \quad \text{(Bahdanau et al., 2014)} \\ \frac{h_t^T \tilde{h_s}}{\sqrt{n}} & \text{Scale Dot-Product} \quad \text{(Vaswani et al., 2017)} \end{cases}$$

Where $h_t$ is the current target state and $\tilde{b_s}$ all the source state. Also in the **General** and **Concatenation** score function $W_a$ and $V_a$ are learnable weight matrices.

**Attention Scoring:**

To improve the Bahdanau attention mechanism, Luong developed two attention models, namely: **Global attention** and **Local attention** model (Luong et al., 2015). We applied the global attention (General, in above equation) for our score calculation.

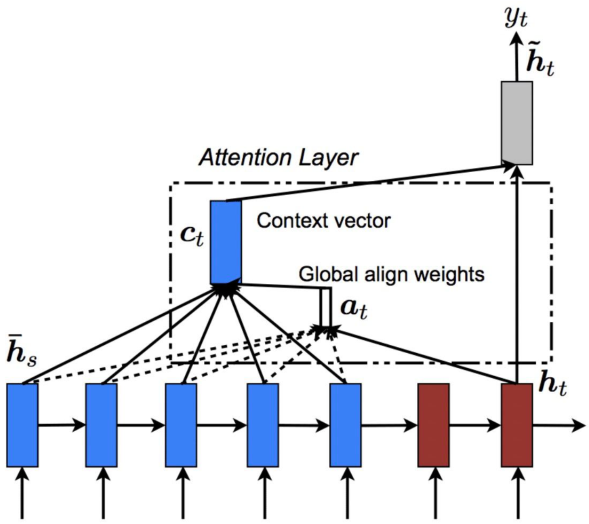



Figure 11: Global Attention in an Encoder-Decoder RNN (Luong et al., 2015)

This approach takes into account all the input sentences and utilizes information from each of the encoder's hidden states to produce each target vector. This is done by applying the **Softmax** activation function to normalize the alignment scores. By normalizing the scores, they could be considered as probabilities, indicating the chances of each encoded input time step or annotation $(h_1, \ldots, h_t)$ being related to the current output time step.

$$\mathbf{a_t} = \text{Softmax}(a_t)$$

This attention vector is then used to weight the encoded representation $h_j^{\text{encoder}}$ to compute the context vector $C_t$, from which the current output time step can be decoded, as defiend above. The context vector $C_t$ is then used along with the decoder hidden state $h_t^{\text{decoder}}$ to compute $\tilde{h_t}^{\text{decoder}}$ which takes $h_t^{\text{decoder}}$ as the outputted hidden state. Now, we concatenate $C_t$ and $h_t^{\text{decoder}}$ when calculating the Softmax distribution over the next word.

$$\hat{y_t} = \text{Softmax}\left(W_{\tilde{h}s}\left[h_t^{\text{decoder}}; C_t\right] + b_s\right)$$

### 3.3.4 Activation Function

We experiment with the different Activation functions to regulate and normalize the given input data via gradient computing.

**The Sigmoid Activation Function** is a non-linear function that composes the input values into the range of 0 to 1.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

**The Hyperbolic Tangent** (tanh) **function** compacts the input values into values between –1 and 1.

$$\sigma(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

**The Softmax Activation function** is used to normalize the output of the networks to a probability distribution over some predicted outputs.

$$\sigma(x)_i = \frac{e^{x_i}}{\sum_{j=1}^{k} e^{x_j}}$$

Where, $\forall i = 1, \ldots, k$ and $x = (x_1, \ldots, x_k) \in \mathbb{R}^k$.

These functions were implemented in the encoder LSTM layer, the decoder LSTM layer, and in the attention layer for computing the attention weights.

### 3.3.5 Loss Function

To train the model, we used the sparse SoftMax cross entropy loss function (SoftMax loss) from the TensorFlow library to calculate the error and minimize the loss function during back-propagation. Although categorical cross entropy loss (CE) eqn has proven effective in complex multiclass classification scenarios, it often suffers from class imbalance problems (Chowdhury and Vig, 2018).

Therefore, we used the SoftMax-cross-entropy loss, which is arguably the most compatible for classification loss due to its simple formulation and probabilistic interpretation (Kobayashi, 2019). Given the logit vectors $\mathbf{f} \in \mathbb{R}^C$ and the label (target seqence) $y = \{1, \ldots, y_c\}$, the SoftMax loss is computed as the cross entropy between the softmax updated probability ($P_y(\mathbf{f})$) and the target (ground truth)

$$P_y(\mathbf{f}) = \frac{e^{f_y}}{\sum_{j=1}^{C} e^{f_j}}$$

$$\mathcal{L}(y, f) = -\sum_{i=1}^{C} y_i \log P_y(\mathbf{f})$$

### 3.3.6 Optimization and Learning rate

In addition to data processing, normalization and backpropagation algorithms minimize the training loss of the network. We adopted the **Adam** optimizer algorithm with a learning rate of 0.001 (Kingma and Ba, 2014). Adam combines the best properties of the AdaGrad and RMSProp optimization algorithms in order to manage sparse gradients on noisy data, and it naturally performs a form of step-size annealing.

## 3.4 Training and Inference Phase

### 3.4.1 Training Steps

To implement the training phase of our seq2seq model, I constructed the customized training loop function and created the model's forward and backward passes. The English input was passed into the encoder layer to generate the hidden and cell states. These hidden and cell states, together with the Igbo inputs, are then supplied into the decoder. I used the teacher-forcing method to train the model accordingly. The teacher enforces the decoder to condition itself to predict the next token according to the given input target.

Furthermore, I estimated the loss by comparing the expected decoder output to the actual values (target). With regard to the encoder and decoder trainable variables, I used the gradient tape technique **tensorflow.GradientTape()** to calculate the gradient.

Subsequently, I executed the training cycle for a predetermined epoch. This iterates over the training dataset, generating the decoder output, calculate the gradient, estimating the BLEU score at every training step (epochs) and the model's parameters were updated accordingly.

### 3.4.2 Testing Phase

Having trained our NMT model, it should be able to translate or decode unseen input sentences. This section of our model is designed to generate significant predictions.

The testing process, is constructed as a method in our model class. This takes supplied input sentences and transforms them into numerical representations using the tokenizer defined during the "data pre-processing phase." The tokens are padded, and the start of sentence and end of sentence identifiers are added accordingly. As previously described, the input vectors are passed to the encoder to retrieve the encoder outputs (hidden state and cell states) and the inputs are initialized as tensors. Unlike the training phase, teacher-forcing is not applied. The decoder gets the previously decoded result to predict the next output.

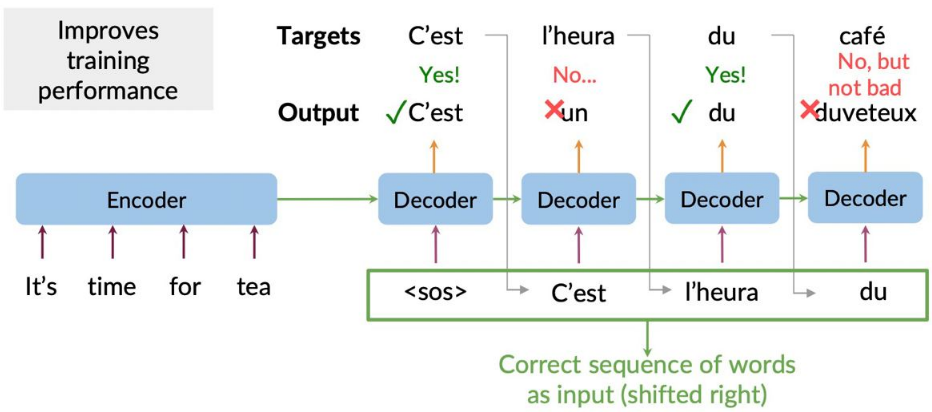



Figure 12: An Example of Teacher Forcing on Engish-to-French Translation (Deeplearing.AI, 2021)

In other words, the training decoder and the inference decoder share comment parameters independently, except for the teacher-forcing technique. Therefore, the probability of the decoder output (predicted output) was calculated using the following techniques.

**Greedy Decoding**

The greedy decoding method is a typical decoding approach for the seq2seq model proposed by Germann (2003), in which the token with the highest conditional probability is generated. The formula is given below.

$$\hat{y}_t = \underset{y}{\operatorname{argmax}} P(y \mid x) = \underset{y}{\operatorname{argmax}} \prod_{t=1}^{T_y} P\left(y^{<t>} \mid x, y^{<1>}, \ldots, y^{<t-1>}\right)$$

**Beam Search Decoding** The beam search algorithm, introduced by Raj Reddy in 1977, is a popular heuristic search algorithm that improves on greedy decoding. The beam search uses graph structure by extending the most probable node. It uses the bread-first-search algorithm to build its search tree. This allows us to find the best sequences over a fixed window size known as the beam width.

At each time step, the beam search generates the predicted word by word from left to right while maintaining a fixed number (beam) of current candidates. The translation performance can be enhanced by increasing the beam size (Freitag and Al-Onaizan, 2017).

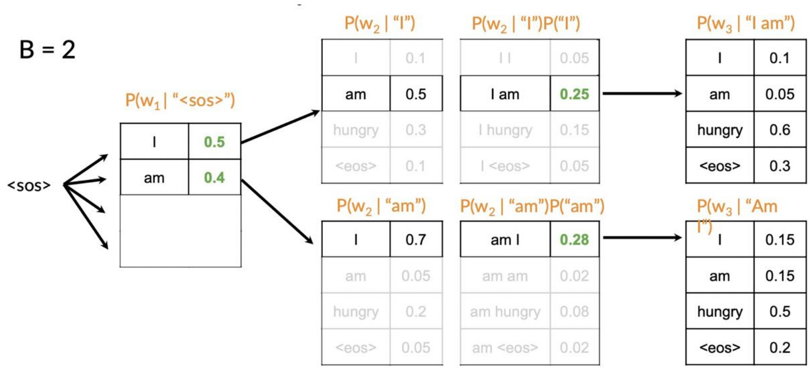



Figure 13: An Example of Beam Search with Beam size = 2 (Deeplearing.AI, 2021)

## 3.5 Evaluation of the Model

We evaluated our model's performance using the Bilingual Evaluation Understudy (BLEU) metric (Papineni et al., 2002), on our dataset (English to Igbo) test set, which has 300 sentences. The BLEU score compares the candidate translation of text to reference (human) translations. Its metric ranges from 0 to 1 (with better score closer to 1). Furthermore, to check the performance of our model with previous work on neural machine translation systems and on a dataset with closer sentence structures, we used the English to French dataset from Tatoeba compilations.

# Chapter 4

## RESULT AND DISCUSSIONS

## 4.1 Introduction

In this section, extensive analysis was conducted to better understand the models' performance. We compared the LSTM and GRU RNN architectures and the impact of different attention scoring functions. We also study the effect of hyperparameter tuning, and decoding techniques on translation quality. Furthermore, we evaluate the general effectiveness of my model on data sets with a closer sentence structure (English-French) and the capacity to handle strange and complex texts.

## 4.2 Model Performance and Results

### 4.2.1 LSTM vs GRU

We experiment with the LSTM and GRU recurrent neural networks in the encoder and decoder layers of our seq2seq model. Table 4.1 shows the parameters of each RNN model. Figure 14 shows the loss and blue score performance of the LSTM against the GRU architecture, after 80 execution epochs.

Furthermore, Table 4.2 and Table 4.3 shows the samples of translation sentences using the LSTM and GRU, respectively. In Figure 14, the loss performance of LSTM appears to be 56.71% superior to that of GRU. Whereas, the GRU model trained 69% faster than the LSTM model and outperformed it with a 2.8% higher BLEU score.

However, from Table 4.2 and Table 4.3, LSTM models gives better performance compares to the GRUs. And considering the nature of our dataset with long sentences, our final model adopts the LSTM recurrent neural networks.

**Table containing lstm and GRU parameters**

| | **LSTM** | **GRU** |
|---|---|---|
| Units | 1024 | 1024 |
| Embedding dim | 256 | 256 |
| Batch size | 128 | 128 |
| Epochs | 80 | 80 |
| Exec. time per Epoch | $\sim$ 3.88 mins | $\sim$ 1.88 mins |
| Total Exec. time | $\sim$ 5.25 hrs | $\sim$ 2.53 hrs |
| Mean loss | 0.0144 | 0.0258 |
| Bleu score | 0.35 | 0.36 |

Table 4.1: **LSTM vs GRU Architectures** – shown is the performance comparison between the LSTM and GRU. The GRU model trained 69% faster than the LSTM model and outperformed it with a 2.8% higher BLEU score. The LSTM model, on the other hand, performed better, with a total loss of 56.71%.

**LSTM vs GRU loss and bleu scores**

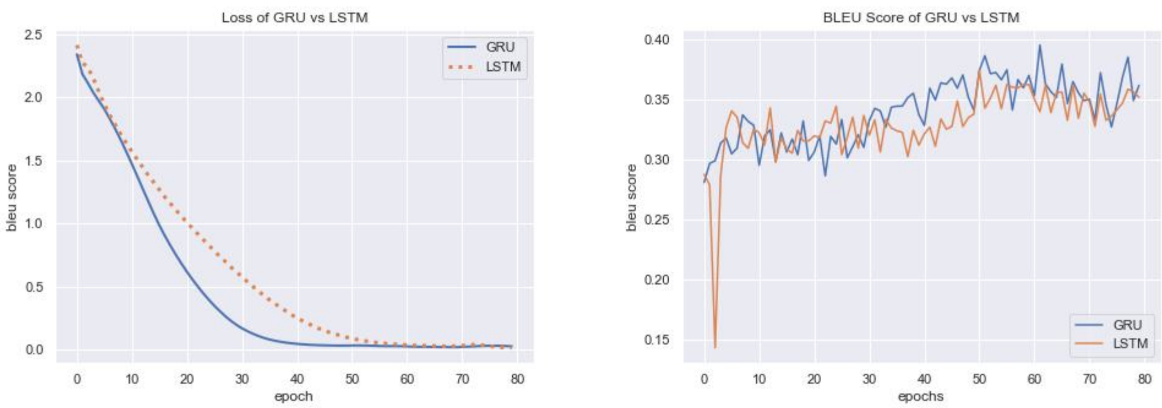



Figure 14: **Loss and BLEU score of the model** – shown are plots of the loss and BLEU score of the GRU model vs LSTM model performance

**English-Igbo Translations**

| | |
|---|---|
| src sentence | interesting news |
| actual translation | akụkọ na-atọ ụtọ |
| Predicted | akụụkọ ndị ga amasị gị **meaning:** *story you like* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | the nigerian government has also asked the south african government to pay for damages done to nigerian businesses in the country. |
| actual translation | Gọọmentị nke Naịjịrịa Gọọmentị nke ọdịda Afịrịka ka ha kwụ ọ ụgwọ ịtịaka n′obi maka mmebị ha mere n′azụmaahịa nke ala Naịjịrịa |
| Predicted | gọọmentị naịjịrịa agwala gọọmentị naịjịrịa agwala gọọmentị naịjịrịa agwala gọọmentị swịss weghachitere ga eme. **meaning:** *The Nigerian government has told the Nigerian government that the Swiss government has returned.* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | super eagles is interesting. |
| actual translation | super ugo na-adọrọ mmasị. |
| Predicted | ndị ọtu super eagles. **meaning:** *members of the super eagles* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | no money was brought for handover in imo state. |
| actual translation | Enweghị ego enyere maka mbufe na agụmakwụkwọ steeti. |
| Predicted | ewepụtaghị ego maka nyefe ọchịchị imo steeti **meaning:** *has not set aside funds for the transfer of state government* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | commissioner for education mr. Maazi Oladoyin Folorunsho, gave the order while inspecting the school to assess the level of damage by the students. |
| actual translation | Kọmishọna nke Agụmakwụkwọ, Maazị Oladoyin Folorunshọ, nyere iwu a mgbe ọ na-eme nnyocha maka ihe niile nke ụmụakwụkwọ ahụ mebiri . |
| Predicted | kọmịshọna na ahụ maka ọrụ maazị marcel ifejiofor toro mbọ nke $o\dot{}\grave{o}tu\dot{}$ ihe owuwu nke ihe gbasara nke mpaghara ahụ bụ ihe gbasara nke mpaghara ahụ bụ ihe gbasara nke mpaghara ahụ bụ ihe gbasara nke mpaghara ahụ bụ ihe gbasara nke mpaghara ahụ bụ ihe gbasara nke mpaghara ahụ bụ ihe gbasara nke mpaghara ahụ bụ ihe gbasara nke **meaning:** *Commissioner Marcel Ifejiofor praised the efforts of the Regional Building Committee. is about that area is about that* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | how best do we describe a secret service or its operatives whose threatens national cohesion the nation s constitution and the judiciary democracy and its majestic |
| actual translation | Kedụ ụzọ kasị mma isi kọwa emume nke nzuzo maọbụ ọrụ ya na-eyija mba, iwu ala na ndị omebe iwu, ọchịchị onye kwuọ uche ya na ụzọ omee ya dị mma? |
| Predicted (**bad pred.**) | kedụ ka ọ na eri nri mana onweghi ego ma ọ bụrụ na ọ bụrụ na ọ bụrụ na ọ bụrụ na ọ bụrụ na ọ bụrụ na ọ ọ bụrụ na ọ bụrụ **meaning:** *how is he eating but no money if if if if if if if if if if if if if if if if if if if if if if if if if if if if if if if if and if* |

Table 4.2: LSTM Sample translations

In Table 4.2 and Table 4.3 for each example, we present some translation segments and highlight a few ones in bold and meaning of predicted sentences in *blue*

| | |
|---|---|
| **English-Igbo Translations** | |
| src sentence | she was born in.. |
| actual translation | a mụrụ ya n afọ . |
| Predicted | a mụrụ ya n afọ . *she was born n.* |
| **English-Igbo Translations** | |
| src sentence | pillars Abia warriors have resolve the feud over enaholo |
| actual translation | pillars abịa warriors emeziela ihu mgbarụ dị n etiti ha maka enaholo. |
| Predicted | gpredicted translation: pillars abịa warriors ekpeziela esemokwu banyere enaholo. **meaning:** *pillars Abia warriors have settled the dispute over enaholo.* |
| **English-Igbo Translations** | |
| src sentence | super eagles is interesting |
| actual translation | super ugo na-adọrọ mmasị. |
| Predicted | ụmụnwaanyị ndị super eagles **meaning:** *ladies of super eagles* |
| **English-Igbo Translations** | |
| src sentence | he now holds the position of minister of state agriculture. |
| actual translation | ugbua o ji ọkwa minista na ahụ maka ọrụ ugbo nke steeti. |
| Predicted | o jizi ọkwa dịka ọtụtụ ndị ndọrọndọrọ ọchịchị dịka ọtụtụ ndị ndọrọndọrọ... **meaning:** *he holds the position of as many politicians as most politicians* |
| **English-Igbo Translations** | |
| src sentence | important news ekweremadu wants equality in government position distribution |
| actual translation | akụkọ ndị dị mkpa ekweremadu chọrọ nhatanha na nkesa ọkwa gọọmentị |
| Predicted (**started well..**) | oakụkọ dị mkpa nsogbu ... **meaning:** *important news problem...* |
| **English-Igbo Translations** | |
| src sentence | he now holds the position of minister of state agriculture. |
| actual translation | ugbua o ji ọkwa minista na ahụ maka ọrụ ugbo nke steeti. |
| Predicted (**performed halfway**) | o jizi ọkwa dịka ọtụtụ ndị ndọrọndọrọ ọchịchị dịka ọtụtụ... **meaning:** *he holds the status of a majority of politicians* |
| **English-Igbo Translations** | |
| src sentence | california fire |
| actual translation | ọkụ kalifornia |
| Predicted (**performed well**) | ọkụ kalifonia **meaning:** *Califonia fire* |
| **English-Igbo Translations** | |
| src sentence | she said for almost five years after i was injured and spent my entire earnings on treating myself i have had no job. |
| actual translation | o kwuru n ihe dịka afọ ise m meruchara ahụ ma mefusie ego niile m nwetara n ịgwọ onwe m enweghị m ọrụ. |
| Predicted (**started well**) | o kwuru ihe m otu afọ asatọ ihe karịrị ọñụ ụwa niile m kpọrọ ndị enyi m ụwa niile ma ihe ndị ihe m otu m gachara n ụlọ bụ nna m ụwa niile m kpọrọ ndị enyi m ... **meaning:** *He said I was eight years old. More than the joy of the world. ...* |

Table 4.3: GRU Sample translations

### 4.2.2 Choices of Attention Scoring

We experiment with the Luong's global attention model and examine the performance of different alignment (scoring) functions; the Bahdanua's concatenation function, the Dot-product and general scoring functions (Luong et al., 2015; Bahdanau et al., 2014).

I present the loss and BLEU score comparisons in the figure (blue and loss). Furthermore, I visualize the alignment weights generated by the respective attention scoring functions used.

| | **Concatenation** | **Dot-product** | **General** |
|---|---|---|---|
| Execution time per Epoch | $\sim$ 4 mins | $\sim$ 2 mins | $\sim$ 3.8 |
| Total Execution time | $\sim$ 5.64 hrs | $\sim$ 3 hrs | $\sim$ 5 hrs |
| Mean loss | 0.0303 | 0.1010 | 0.8421 |
| Bleu score | 0.358 | 0.36 | 0.315 |

Table 4.4: **Performance of different Scoring Function** – when compared to the **concat** and **general**, the LSTM +**dot** model trained 50% faster and outperformed them with 2.81% and 13.33% BLEU, respectively.

**Loss and Bleu score comparison of different attention scoring functions**

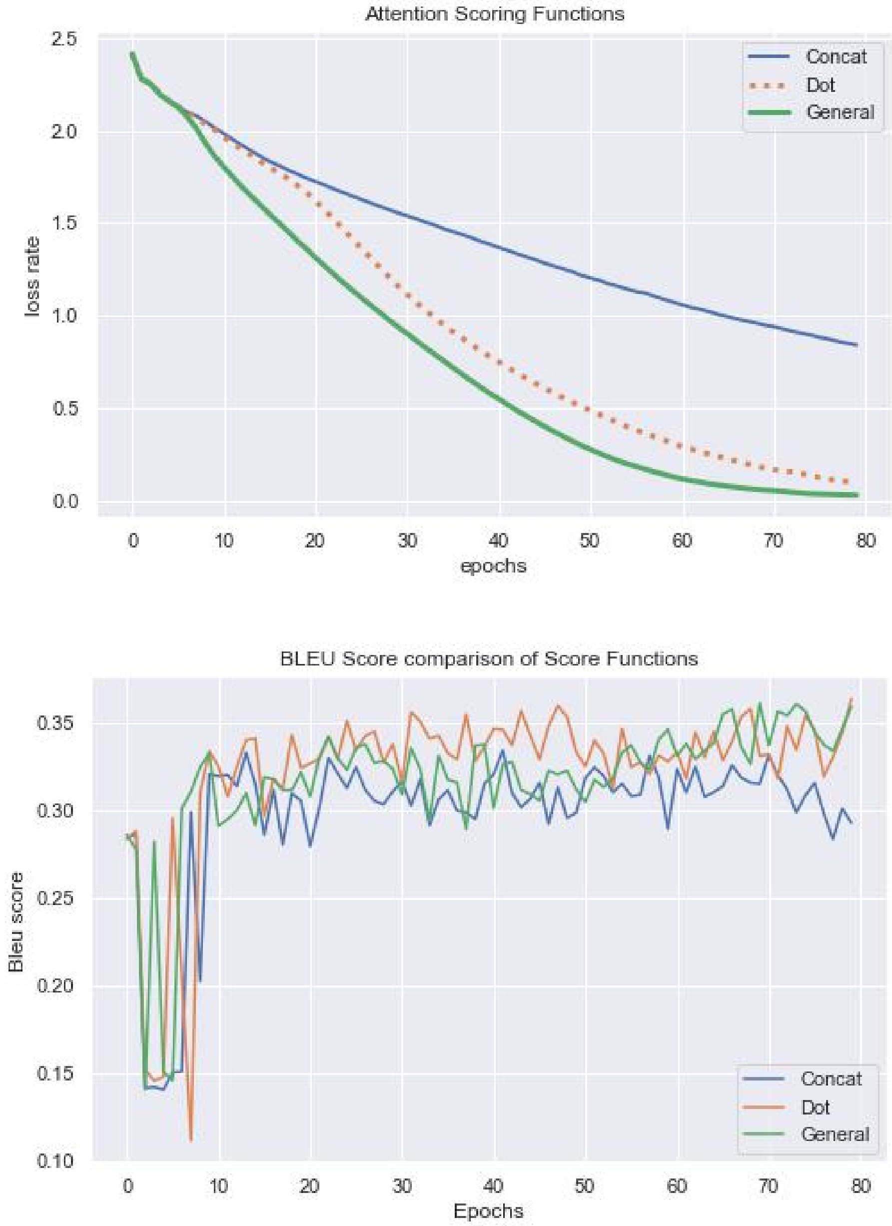



Figure 15: **Loss and BLEU score of the model** – shown are plots of the loss and BLEU score of different Global attention scoring function along with the LSTM model performance

**English-Igbo Translations**

| | |
|---|---|
| src sentence | interesting news |
| actual translation | akụkọ na-atọ ụtọ |
| Predicted **fairly close** | akụkọ ndị ga amasị gị **meaning:** *story you like* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | Peter Afunanya is a poor image maker for the DSS with his defence and blatant falsehood he puts forward on the several viral videos that the desecration of the hall of justice at the federal high court Abuja on Friday by the DSS. |
| actual translation | na mwute, Peter Afunanya bụ onye mmetọ aha nke DSS site na nchekwa iberibe ya na nnukwu asị nke o tinyeworo n′ihe ngosi onyoonyoo dị iche nke gosipụtara mmetọ nyere ụlọ ọgbakọ nke ikpe na ụlọikpe dị Elu nke Etiti, dị n' Abụja, n'ụbọchị Fụraịde site n'aka ndị DSS. |
| Predicted **poorly** | otinye ego n ihi okwu na onye ntaakụkọ sahara reports na onye ntaakụkọ... . **meaning:** *invests due to issues with sahara reporter and sahara reporter reporter sahara...* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | super eagles is interesting. |
| actual translation | super ugo na-adọrọ mmasị. |
| Predicted **poorly** | akụkọ ndị ga amasị gị **meaning:** *stories you like* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | no money was brought for handover in imo state. |
| actual translation | Enweghị ego enyere maka mbufe na agụmakwụkwọ steeti. |
| Predicted **fairly close** | ewepụtaghị ego n imo steeti **meaning:** *no budget for state education* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | commissioner for education mr. Maazi Oladoyin Folorunsho, gave the order while inspecting the school to assess the level of damage by the students. |
| actual translation | Kọmishọna nke Agụmakwụkwọ, Maazị Oladoyin Folorunshọ, nyere iwu a mgbe ọ na-eme nnyocha maka ihe niile nke ụmụakwụkwọ ahụ mebiri . |
| Predicted **fair** | gọvanọ nke a ga eme ka ọ bụ mazi uchenna okafor kwuru na ndị otu ahụ kwuru na ndị otu ahụ kwuru .... **meaning:** *this governor will be made by Mr. uchenna okafor said the party members said the party members said...* |

**English-Igbo Translations**

| | |
|---|---|
| src sentence | one of the greatest dangers for any business is the violation of the rule of law and the principles of justice equality and the right to be heard. |
| actual translation | Otu n'ime ihe kasị dị egwu maka azụmahịa ọbụla nke na-ada iwu ọchịchị na usoro ikpe mkwụmọtọ, ịha ụha na ikike ịnụ olu |
| Predicted (**poorly**) | nchọcha a ga esi pụta n ihi gịnị? **meaning:** *What will be the result of this research?* |

Table 4.5: General Scoring function Sample translations

| English-Igbo Translations | |
|---|---|
| src sentence | important news today |
| actual translation | akụkọ na-atọ ụtọ |
| Predicted **good** | akụụkọ dị mkpa taa **meaning:** *important news today* |
| **English-Igbo Translations** | |
| src sentence | no money was brought for handover in imo state. |
| actual translation | Enweghị ego enyere maka mbufe na agụmakwụkwọ steeti. |
| Predicted | ewepụtaghị ego maka nyefe ọchịchị imo steeti **meaning:** *has not set aside funds for the transfer of state government* |
| **English-Igbo Translations** | |
| src sentence | participants at a two day capacity training workshop in abakaliki ebonyi state have identified the deconstruction of patriarchal dominant view in society as a major step towards actualization of thirty five per cent affirmative action. |
| actual translation | ndị sonyere na nkụzị ụbọchị abụọ n abakiliki ebonyi steeti achọpụtala nruzighari nke echiche ndị isi ala na ọha mmadụ dịka nnukwu nzọụkwụ iji gosipụta mmezu nkwupụta pesent iri atọ na ise. |
| Predicted **good on long sen.** | ndị sonyere na nkụzị ụbọchị abụọ n abakiliki ebonyi steeti achọpụtala nruzighari nke echiche ndị isi ala na ọha mmadụ dị ka nnukwu nzọụkwụ iji gosipụta mmezu nkwupụta iri atọ na ise. **meaning:** *Participants in the two-day program in Abakiliki ebonyi state have identified the reversal of the views of local leaders and society as a major step towards demonstrating the implementation of the 35 resolutions.* |
| **English-Igbo Translations** | |
| src sentence | the warriors who will compete in Imo and Abia states |
| actual translation | dike ndị ga akwata ya n imo na abia steeti |
| Predicted (**very good.**) | dike ndị ga akwata ya n imo na abia steeti **meaning:** *the warriors who will compete in Imo and Abia states* |
| **English-Igbo Translations** | |
| src sentence | He said that the law if enacted would protect respect and promote the rights of women and children in order to improve their health and their standard of living. |
| actual translation | o kwuru na ọ bụrụ na e mebe iwu ahụ na ọ ga echekwa sopụrụ ma kwalite ikike dịịrị ụmụnwaanyị na ụmụaka iji bulite ahụ ike ha nakwa obibi ndụ ha. |
| Predicted (**very good.**) | dike ndị ga akwata ya n imo na abia steeti **meaning:** *she said the law and its construction will protect and promote women's rights and its construction will protect and promote women's rights...* |

Table 4.6: Dot-Product Sample translations

In Table 4.5 and Table 4.6 for each example, we present some translation segments and highlight a few ones in bold and the meaning of predicted sentences in *blue*

| English-Igbo Translations | |
|---|---|
| src sentence | interesting news |
| actual translation | akụkọ na-atọ ụtọ |
| Predicted **Greedy algorithm** | akụkọ ndị ga amasị gị **meaning:** *story you like* |
| Predicted **Beam search** | akụkọ dị mkpa **meaning:** *interesting story* |

Table 4.7: Sample translations of Beam Search and Greedy algorithm

### 4.2.3 Decoding Techniques

We experiment with the "greedy decoding" technique (Germann, 2003) and the Beam search decoding technique. Table 4.7, illustrates the performance of both methods. My seq2seq model turns out to translate effectively on greedy decoding on long sentences as experimented by Sutskever et al. (2014) .

### 4.2.4 Error Minimization and Loss Function per Epoch

We evaluate the error minimization of my model for in each epoch in the training process, Figure 16 illustrates the decreases in error rates for the first 50 iterations and the next 50 iterations.

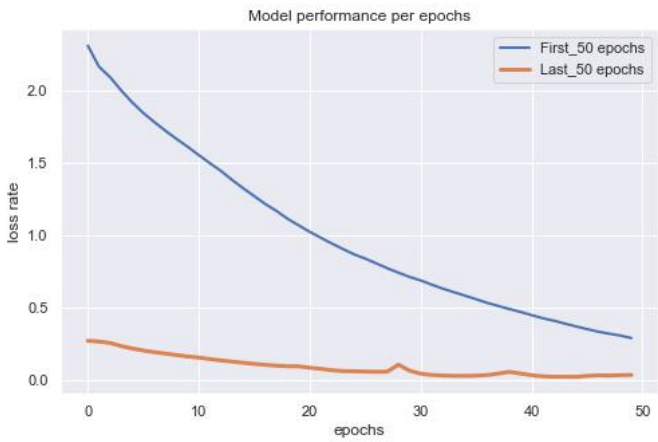



Figure 16: **Model performance per epochs** – shown is the error decrease for 100 epochs

| | **Mean Error** | **Bleu score** |
|---|---|---|
| First 50 Epochs | 0.2727 | 0.3303 |
| Next 50 Epochs | 0.0143 | 0.3817 |

Table 4.8: Performance evaluation per Epochs

From Table 4.8 , the mean error for the first 50 epochs is approximately 0.2727. And it drops dramatically to 0.0143 over the next 50 epochs, which is a decent performance.

### 4.2.5 Hyper-Parameter Tuning

**Effect of Batch sizes on Model Performance**

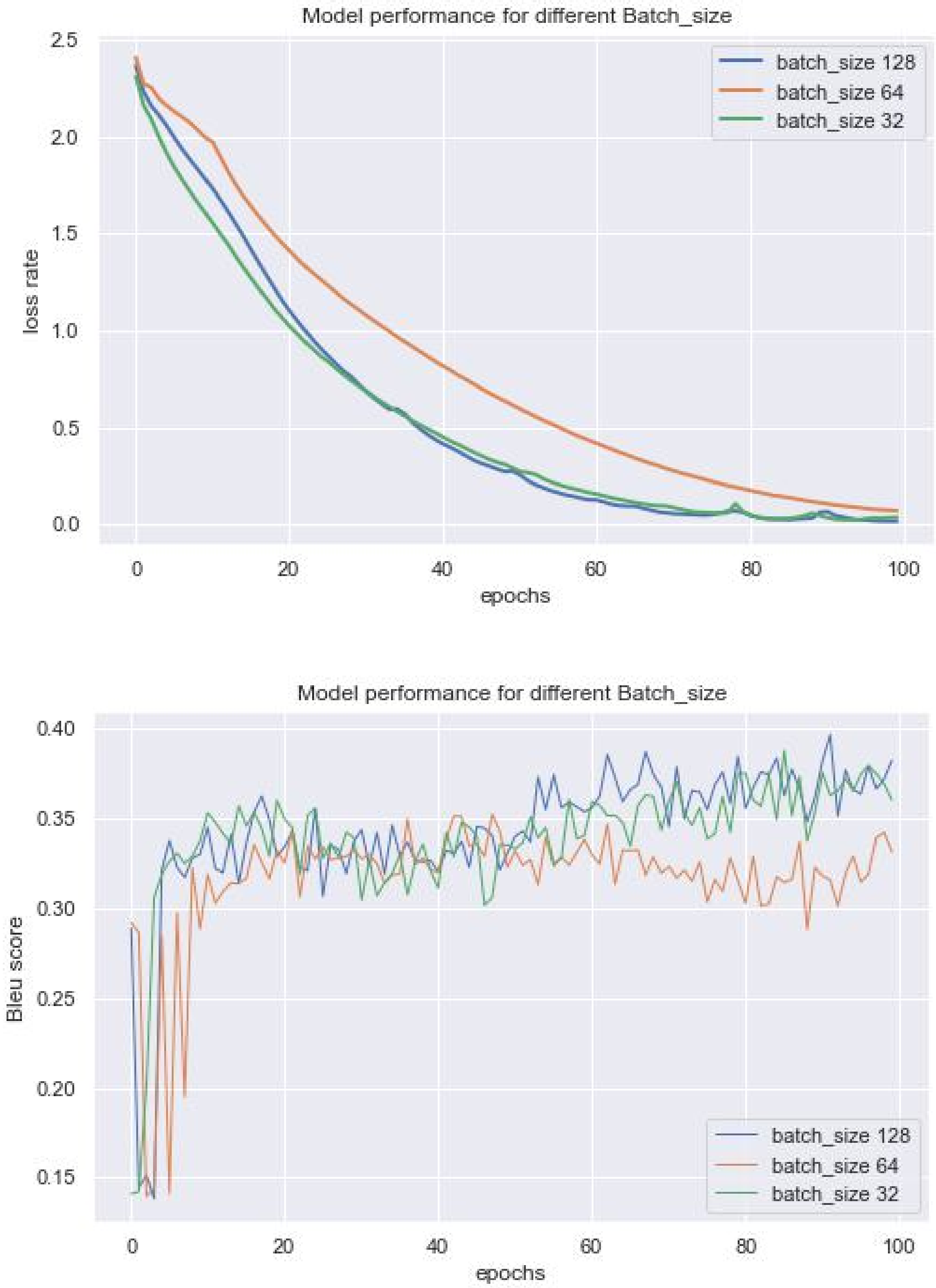
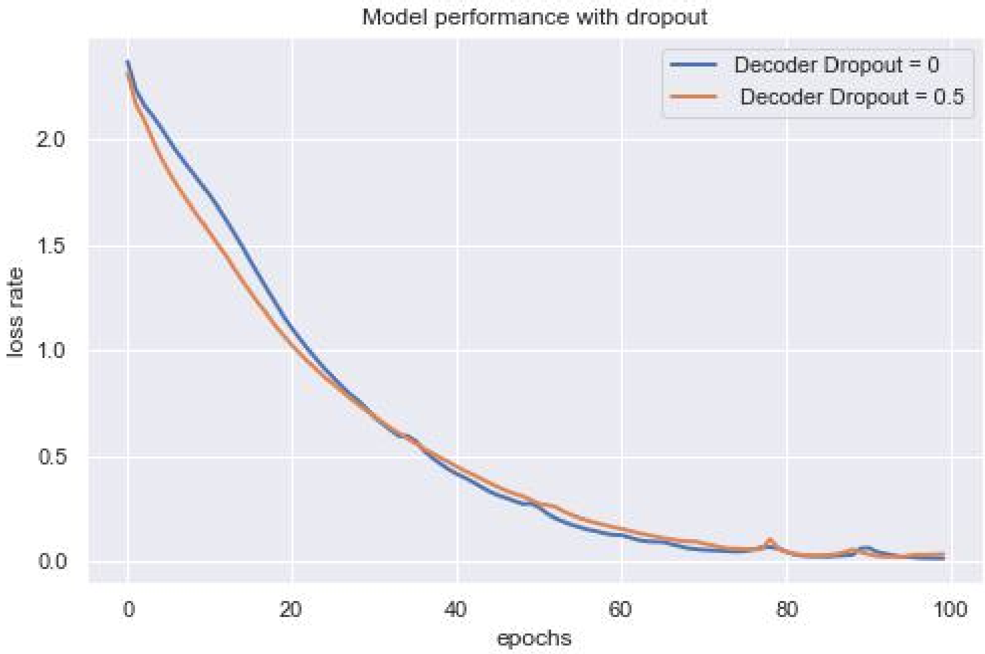



Figure 17: **Loss and BLEU score for different Batch sizes** – shown is the loss error and BLEU score performance of our model with different batch sizes.

**Effect of Dropout on Model Performance**



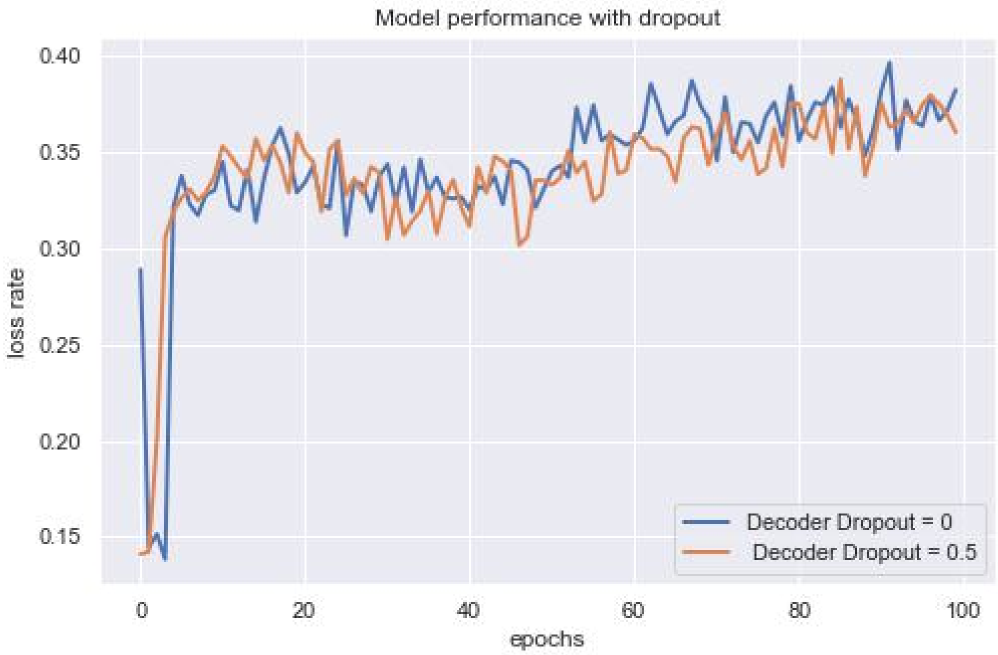
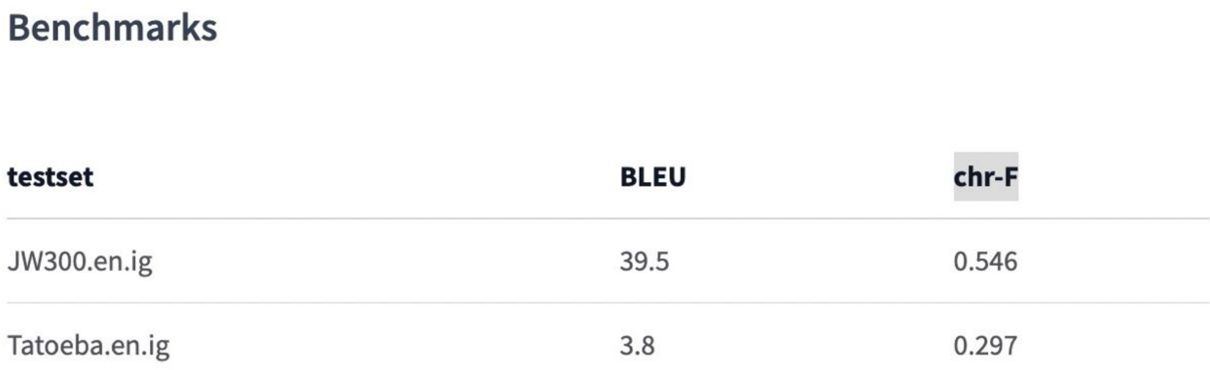




Figure 18: **Loss and BLEU score for different Dropout** – shown is the loss error and BLEU score performance of our model with different Dropout.

| parameters | values |
|---|---|
| Epochs | 100 |
| LSTM unit | 1024 |
| Batch-size | 32 |
| Decoder dropout | 0.5 |
| total loss | 0.0143 |
| Bleu score | 0.3817 |

Table 4.9: **Performance of the Developed model** – the developed **LSTM + dot** model gave the optimal performance of 0.0143 total loss and a 0.3817 BLUE score, this is a impressive result compared to the **Tatoeba.en.ig and JW300.en.ig benchmarks**


**Benchmarks**

| testset | BLEU | chr-F |
|---|---|---|
| JW300.en.ig | 39.5 | 0.546 |
| Tatoeba.en.ig | 3.8 | 0.297 |

Figure 19: **Benchmarks for English-Igbo dataset** – shown is the state-of-the-art for **Tatoeba.en.ig and JW300.en.ig provided by Helsinki-NLP/OPUS-MT-train.**

## 4.3 Model Performance on Datasets with closer Sentence Structure

Furthermore, we examine the overall performance of my seq2seq model on a simpler dataset (English to French parallel corpus with ”subject-verb-object” (SVO) word order). Due to limitation in computational space, we used approximately 70,000 sentence pairs from the Tatoeba compilation.

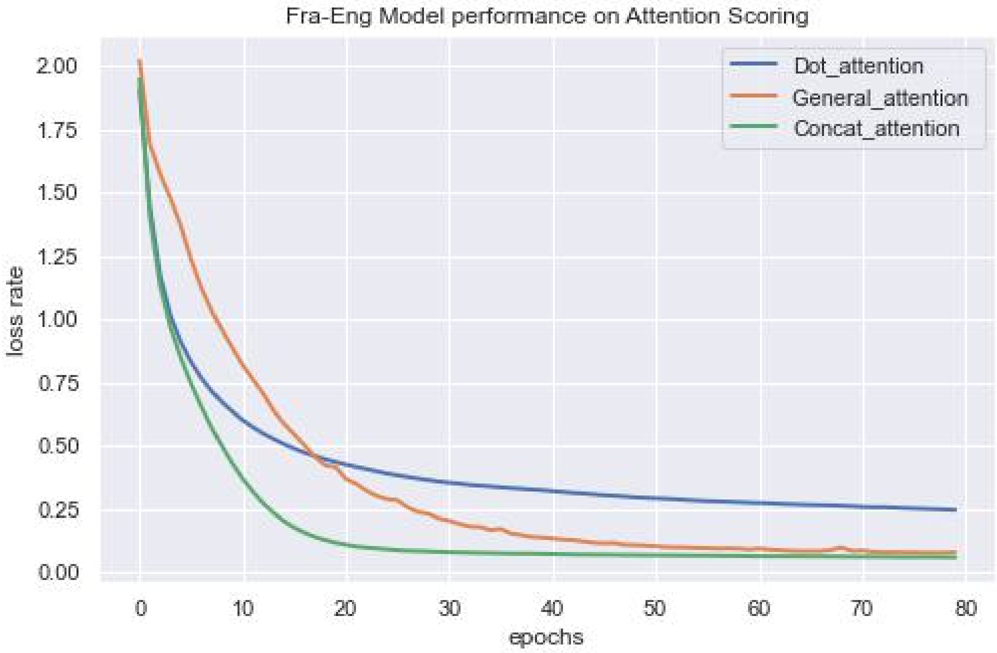
#### Performance with different attention scoring with Fra-Eng

After training the model for 80 epochs with different attention scoring functions, the concatenation and general attention scoring gave higher BLEU score of 0.59, 0.587 and the total loss of 0.0620, 0.0818. It is interesting to observe that that Dot-product is 82% faster than the General attention and 56% faster than the Concat attention. Whereas, compared to these two scoring functions, the Dot attention does not yield good results on the Tatoeba Eng-Fra data sets, further analysis should be conducted to investigate this reason.

Table 4.10 shows the results of experimentation with $70,000$ sentence pairs. Our seq2seq model with the concat, dot and general attention scoring provides the BLEU scores of 0.59, 0.52 and 0.587 which improve on the Tatoeba English-French benchmark of 0.505 by a score of "9" provided by Helsinki-NLP (Tatoaba, 2022)

| | **concat** | **Dot-product** | **General** |
|---|---|---|---|
| Time per Epoch | $\sim$ 1.76 mins | $\sim$ 1.33 mins | $\sim$ 3.166 mins |
| Total Exec. time | $\sim$ 2.36 hrs | $\sim$ 1.80 hrs | $\sim$ 4.304 hrs |
| Mean loss | 0.0620 | 0.2480 | 0.0818 |
| BLEU score | 0.59 | 0.526 | 0.587 |

Table 4.10: **Performance of My Model with Fra-Eng Dataset** – Shown is the performance of My Model on the English to French dataset with different scoring functions. The **concat** and **dot** gave the best BLEU score and loss performance



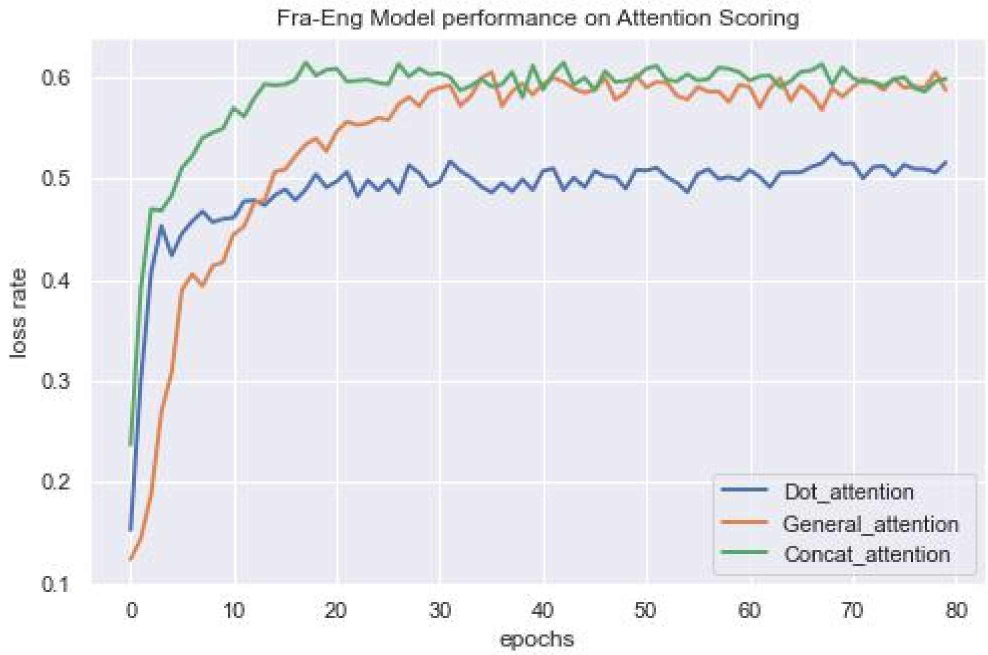




Figure 20: **Loss and BLEU score with Fra-Eng Dataset** – shown are plots of the Loss and BLEU score performance of Fra-Eng Dataset with different scoring functions

## 4.4 Performance and Effect of Transfer Learning

### 4.4.1 SimpleTransformer.Seq2Seq Model (MarianNMT)

It has been demonstrated that transfer learning improves translation quality for low-resource languages (Zoph et al., 2016; Nguyen and Chiang, 2017; Nair et al., 2022). Hence, we apply the transfer learning method to improve the performance of our translation. I experiment with the Simple Transformer library by HuggingFace (Rajapakse, 2020; Wolf et al., 2019).

Hugging-Face is an open-source platform that offers machine learning models and datasets, as well as dozens of pre-trained transformer models for developing a variety of NLP tasks such as language generation,machine translation, question answering, seq2seq tasks, Name Entity Recognition, and conversational AI. Each of these tasks has its own pre-trained Simple Transformers model, allowing the models to be fine-tuned to their unique use case.

From the simple-transformer compilation, we utilize the MarianNMT Fast Neural Machine Translation model built by the Microsoft Translator Team, which uses the OPUS pretrained model data by Jörg Tiedemann. The model consists of deep RNNs, transformer-based models, and models developed in C++ to allow for quick automated differentiation (Junczys-Dowmunt et al., 2018). It supports a variety of languages (including the English-Igbo language pair). The MarianNMT consists of 6-layer blocks of encoder and 6-layers of decoder, and it was trained to synchronous Adam on 8 NVIDIA Titan X Pascal GPUs with 12GB RAM for 7 epochs each (Junczys-Dowmunt et al., 2018), making it easy to build and fast for training and translation.

#### Training and Performance

Training of the model takes 5.13 hours for 20 epochs, with each epoch taking about 15 minutes to complete. This gives a BLEU score performance of 0.43 on 597 test datasets. The performance distribution is shown in Table 4.11 and Figrue 22 22. 27% of the test data produced a score of less than 0.35, whereas 40% fell between the scores of 0.35 and 0.50. Finally, 33% of the test data gives a score greater than 0.50. Therefore, when comparing the average BLEU score of 0.43 with the Hugging-Face benchmark score of 0.38 and 0.395, it is clear that our transfer learning system performs significantly well.

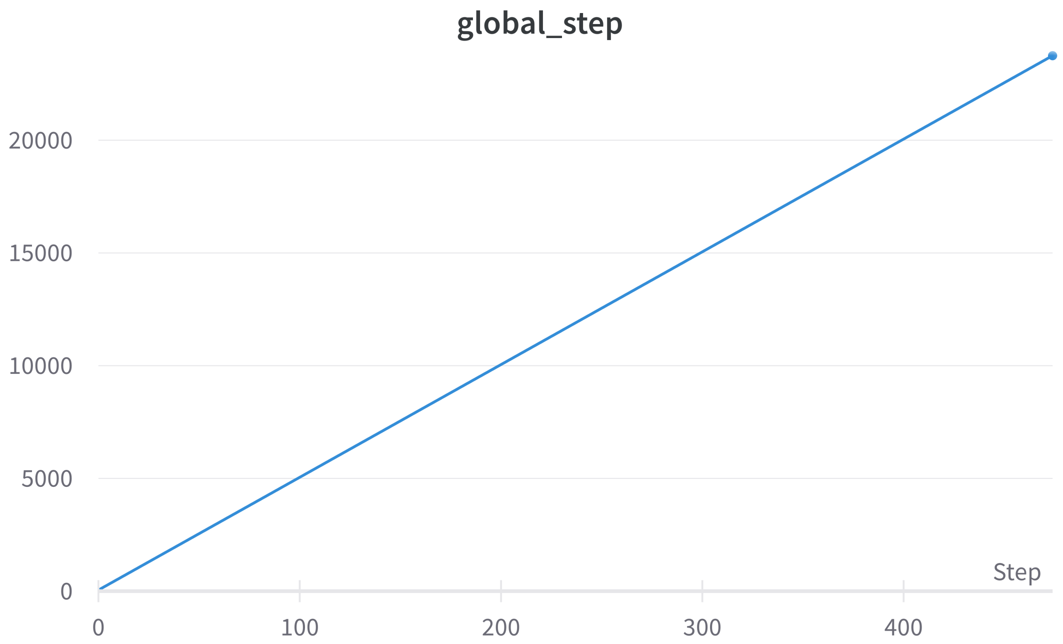




Figure 21: **Graph of training loss and Time taken for training** – shown is the Loss performance of Eng-Igbo Dataset on Seq2Seq Tranformer model

We used the NVIDIA Tesla K80 GPU to train the model, and we automatically save the training weights and biases by setting the Wandb-project value in modelArgs (Biewald, 2020). (**Wandb** is a machine learning model experimental tracking tool.) The total loss value of 0.6020 does not tell us much about the model performance, however, the drastic decrease in loss after 20 epochs indicates that the model is learning fast. But considering GPU memory limitation we could only train for 20 epochs for 5 hours (300 minutes).

The continuous decrease in training loss in Figure 21 indicates that the model has not fully converged. Training for more epochs would improve the performance of the model, but it would need an extra 5 to 10 hours of runtime.

| | eng-ig | BLEU | Vocab size |
|---|---|---|---|
| Benchmark | Tatoaba | 0.38 | 18$k$ |
| | JW300 | 0.395 | 20$k$ |
| My Model | RNN-based | 0.3817 | 16$k$ |
| | Transfer | 0.43 | 16$k$ |

Table 4.11: **My Model vs state-of-art performance** – Whereas the RNN-based model gives a comparable score of 0.3817 compared to the Totoaba benchmark and slightly behind the JW300 benchmark. The transfer learning approach increased the performance by 4.83 points. Hence, with 16K vocab size our system yields a significant improvement over the existing benchmarks provided by Helsinki-NLP/OPUS-MT-train.

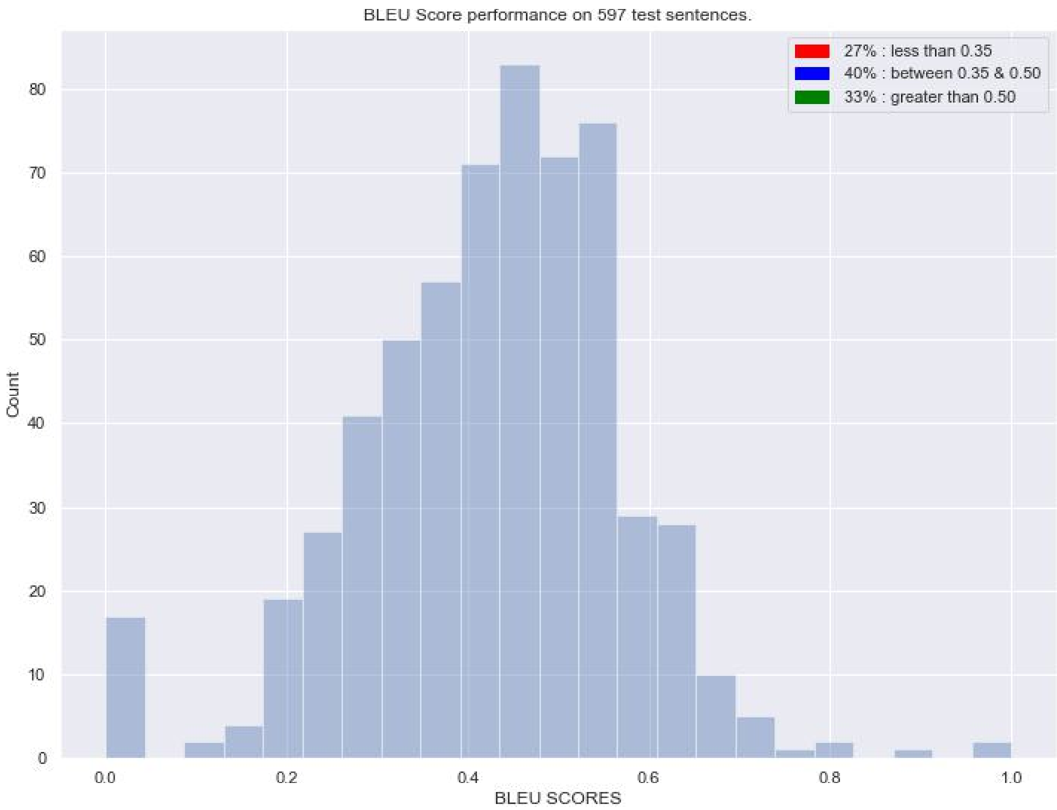



Figure 22: **BLEU Score performance** – BLUE score distribution across 597 test data set on the Seq2Seq Tranformer model

| English-Igbo Translations | |
|---|---|
| src sentence | Several Nigerians had been murdered before those attacks started. |
| actual translation | egbuola ọtụtụ ndị Naijirịa tutu mwakpo ndị ahụ malite |
| Predicted **good** | E gbagburu ọtụtụ ndị Naijirịa tupu mwakpo a-amalite. **meaning:** *Many Nigerians were shot dead before the attack.* |
| **English-Igbo Translations** | |
| src sentence | A Sick Nation And Her Sickening Secret Service By Erasmus Ikhide |
| actual translation | Mba nọ n'ọrịa na emume nzuzo ya ahụ adịghị site n'aka Erasmus Ikhide. |
| Predicted **fair** | Mba na-arịa ọrịa na ọrịa ọrịa bekee site na mmụọnta Ikpu. **meaning:** *A Nation plagued by disease and epidemics by the influence of Ikpu* |
| **English-Igbo Translations** | |
| src sentence *long sentence* | Without that, Nigerians would wake up one day and be told that the notorious scofflaw has created her own law courts where they (the DSS) can now do maximum violence to Nigerian citizens, democracy and civilisation. |
| actual translation | ma ewepụ ya, ụmụ afọ Naijirịa ga-eteta ụra otu ụbọchị ma nụ na ekperima ahụ ewubela ụlọikpe nke aka ya ebe ha (DSS) ga-anọ mebige ndị Naijirịa, ọchịchị onye kwuo uche ya na mmepe obodo. |
| Predicted **good on long sen.** | E nweghị nke ahụ, ndị Naijirịa ga-ebulie otu ụbọchị ụbọchị a ma ama ama emepụtala ụlọikpe nke ya ebe ha (ndị DSS) nwere ike-ime ihe ike-ime ọmụma aghara ndị Naijirịa, ọchịchị onye kwuo uche ya na. **meaning:** *Otherwise, Nigerians will rise to the occasion of a landmark court where they (the DSS) can commit acts of violence against Nigerians, democracy and.* |
| **English-Igbo Translations** | |
| src sentence *special characters* | Follow me on Twitter @ikhide_erasmus1 |
| actual translation | soro m na ọwaozi Twita @ikhide_erasmus1 |
| Predicted (**very good.**) | soro m na Twitter @ikhide_erasmus1 **meaning:** *Follow me on Twitter @ikhide_erasmus1* |
| **English-Igbo Translations** | |
| src sentence | May God bless those in our justice system who stood on the right side of history by doing the right thing. |
| actual translation | Ka Chukwu gọzie ndị niile nọ n'ụlọ omebe iwu anyị ndị nọ n'ụzọ aka nri nke akụkọ site n'ime ihe dị mma. |
| Predicted (**very good.**) | Kpee, Chineke Chineke na-akwado ndị nọ n'-usoro ikpe anyị bụ ndị kwụrụ n'akụkụkụ ihe mere eme site n' ihe ziri ezi. **meaning:** *Pray God, God upholds those in our legal system who stand on the side of history through righteousness.* |
| **English-Igbo Translations** | |
| src sentence | Sudan's Ex-President, Al-Bashir, Sentenced To Two Years For Corruption. |
| actual translation | Onyeisi ala chịrịla Sudan, Al-Bashir, bụ onye amarala ikpe ije mkpọrọ afọ abụọ maka mpụ. |
| Predicted (**good.**) | Onye mpaghara ndị ọchịchị Sudan, Al-Bashir, chọrọ n'ọchịchị afọ abụọ maka nrụrụ. **meaning:** *Al-Bashir is wanted by the Sudanese government for two years in prison for corruption* |

Table 4.12: **Sample translations from Transfer learning** – Here we present some translated samples from the MarianNMT transfer learning model. Highlighted is the meaning of predicted sentences in *blue*. Our model has proven to predict above 70% correct meaning of sentences, which is an outstanding and surprising outcome.

## 4.5 Discussion

### 4.5.1 LSTM vs GRU

The developed seq2seq model can translate a given input English sentence to a parallel Igbo sentence. The NMT model is built with the RNN encoder and decoder. We experiment with both the LSTM and GRU RNNs after 80 epochs, as illustrated in Table 4.1and Figure 14 The LSTM architecture tends to outperform the GRU in terms of loss performance, with a total loss of 0.0144 against the GRU loss of 0.0258. whereas the GRU gave a higher BLEU score of 0.36 against the LSTM score of 0.35.

As shown in Table 4.2, the GRU gives a better performance on short sentences, which confirms the works of Khandelwal et al. (2016). However, the LSTM model in Table 4.3 outperforms the GRU on longer sentences. And considering the sentence length of our parallel corpus, we adopt the LSTM architecture for the encoder and decoder layers.

### 4.5.2 Effect of Attention Scoring

My model also incorporates the attention mechanism proposed by Bahdanau et al. (2014). I experimented with Luang's global attention model (Luong et al., 2015) and different scoring functions (concatenation, Dot-product, and General). The results are illustrated in Table 4.4 and Figure 15. The results reveal that concatenation and Dot-Product scoring functions give better loss performance, whereas the Dot-Product gives a higher BLEU score and translation sample, as illustrated in Table 4.6. Our results correlate with the original work of Luong, Pham, and Manning (2015).

### 4.5.3 Effect of decoding techniques and hyperparameter tuning

We compare beam search and greedy decoding techniques to improve the translation quality of our model. The beam search with a beam size of 5 gives a better translation quality on shorter sentences Table 4.7. This supports the findings of Sutskever et al. (2014). Furthermore, for longer sentences, my model performs well on greedy decoder. Table 4.6.

After several hyperparameter experiments, my model (an LSTM-based model with Dot-product attention) gave the optimal performance of 0.0143 total loss and a 0.3817 BLUE score with the listed parameters: 100 epochs, 32 batch-size, 0.5 decoder dropout, 1024 LSTM units, and 256 Embedding dimension.

Therefore, my developed model ( 0.3817 Bleu score) outperforms the state-of-the-art benchmark ( 0.38 bleu) provided by TatoabaEngIgbo (2020) and has comparable performance with the JW300 benchmark ( 0.395 ) as shown in Table 4.9 and Figure 19.

### 4.5.4 Performance of my model with a simpler data set (Eng-Fra)

To verify the performance of my model on a simpler data set with closer "subject-verb-object" (SVO), I trained my model on 70,000 Taboaba English to French sentence pairs as illustrated in Table 4.10 and Figure 20. The Bahdanua concatenation and general attention scoring gave the best performance of 0.0620, 0.0818 total loss, and 0.59, 0.587 BLEU score, respectively. This shows that our model improved on the Tatoeba English-French data set benchmark of 0.505 by a score of "9" as displayed in Table 4.10.

### 4.5.5 Effect of Transfer Learning

Using the MarianNMT transformer based-model by Junczys-Dowmunt et al. (2018) mentioned in the Section 4.4.1. We improved the translation performance of our model from 0.38 to 0.43 Bleu score. From Table 4.11 and Table 4.12, our transfer learning model outperformed the benchmark provided by Tatoaba and JW300, which is an impressive result.

In other words, despite the limited training data set, our models (RNN-based model with 0.38 bleu and Transfer learning model with 0.43 bleu) outperform and outperform the state-of-the-art benchmarks (0.38 and 0.395) for Eng-Igbo translation. Hence, our research proffers a new possibility to get a well-balanced translation from English to Igbo language and other low-resource languages.

Importantly, our results show that the RNN-based models combined with an attention mechanism create a greater quality of translation than earlier neural machine translators. It was validated by the works of Chen et al. (2018), who examined several NMT systems and found that recurrent neural MT outperformed other systems considered.

# Chapter 5

## CONCLUSIONS

In this research, we developed an RNN-based Neural Machine Translation model from English to Igbo. We have also shown the effect of transfer learning techniques (Zoph et al., 2016; Nguyen and Chiang, 2017) on low-resource languages using the Marian NMT toolkits by (Junczys-Dowmunt et al., 2018).

Our RNN-based developed model provides good translation and comparable results with the Totaoba and JW300 state-of-the-art benchmarks despite our limited training dataset. And when trained on a simpler sentence pair (English-French), our model increased on the Totaoba eng-fra benchmark by a BLEU score of 9.

Furthermore, our transfer learning approach increased the English-Igbo benchmark by 4.83 BLEU scores, achieving a translation quality of approximately 70% , which is a fascinating result considering the complex sentence structure of our parallel corpus. I believe the efficiency with which my model achieves fluent translation could be utilized to construct powerful models capable of translating low-resource languages. By taking into consideration the exact meaning of words with complicated SVO.

## 5.1 Limitation and Future Works

The major limitation we encountered during this research was a lack of GPU memory space to further extend the train depth of our developed model and the transfer learning model. This is expensive to obtain.

In future study, we propose increasing the research's vocabulary size and expanding the model to multilingual Neural Machine translation for low-resource languages.

Furthermore, concerning the model's design, a more advanced attention model could be applied. We used the global attention mechanism in our research. However, local attention may increase performance (Luong et al., 2015). Since the Beam Search decoding technique tends to penalize long sentences, and it is computational expensive and consumes a lot of memory, we recommend future research to implement the Minimum Bayes Risk (MBR) and ROUGE decoding methods.

Finally, we recommend the application of Graph Neural Networks (GNN) to improve the translation quality. In recent studies, GNN has been shown to be effective in solving semantic problems (the study of meaning in language) by incorporating the syntactic graph structure into neural attention-based encoder-decoder models for machine translation. (Bastings et al., 2017; Yin et al., 2020)

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